November/December 2014

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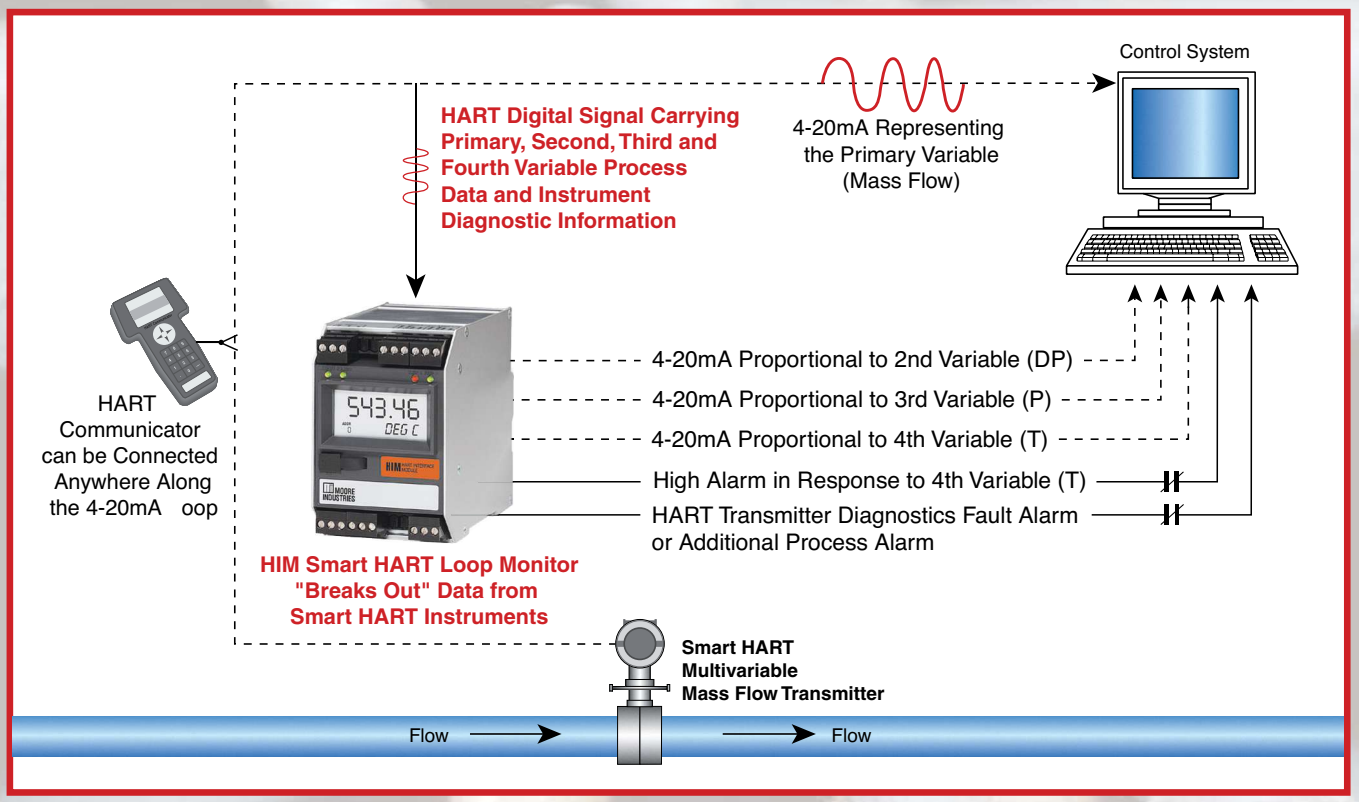
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ISA106 and the importance of automating manual procedures

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Auditing alarm management systems

Alarm management is a company's commitment to safety as a critical focus of operation. This article discusses auditing alarm systems, which is the essence of continuous improvement to maintain and improve management and work practices. Read more at: **www.isa.org/intech201412web01**.

Strong asset management performance

With increasing pressures to optimize costs and reduce the risk of industrial incidents, it is critical that inspection and maintenance efforts be streamlined for utmost efficiency. Establishing a strong asset performance management foundation can generate significant value. Read more at: **www.isa.org/intech201412web02**.

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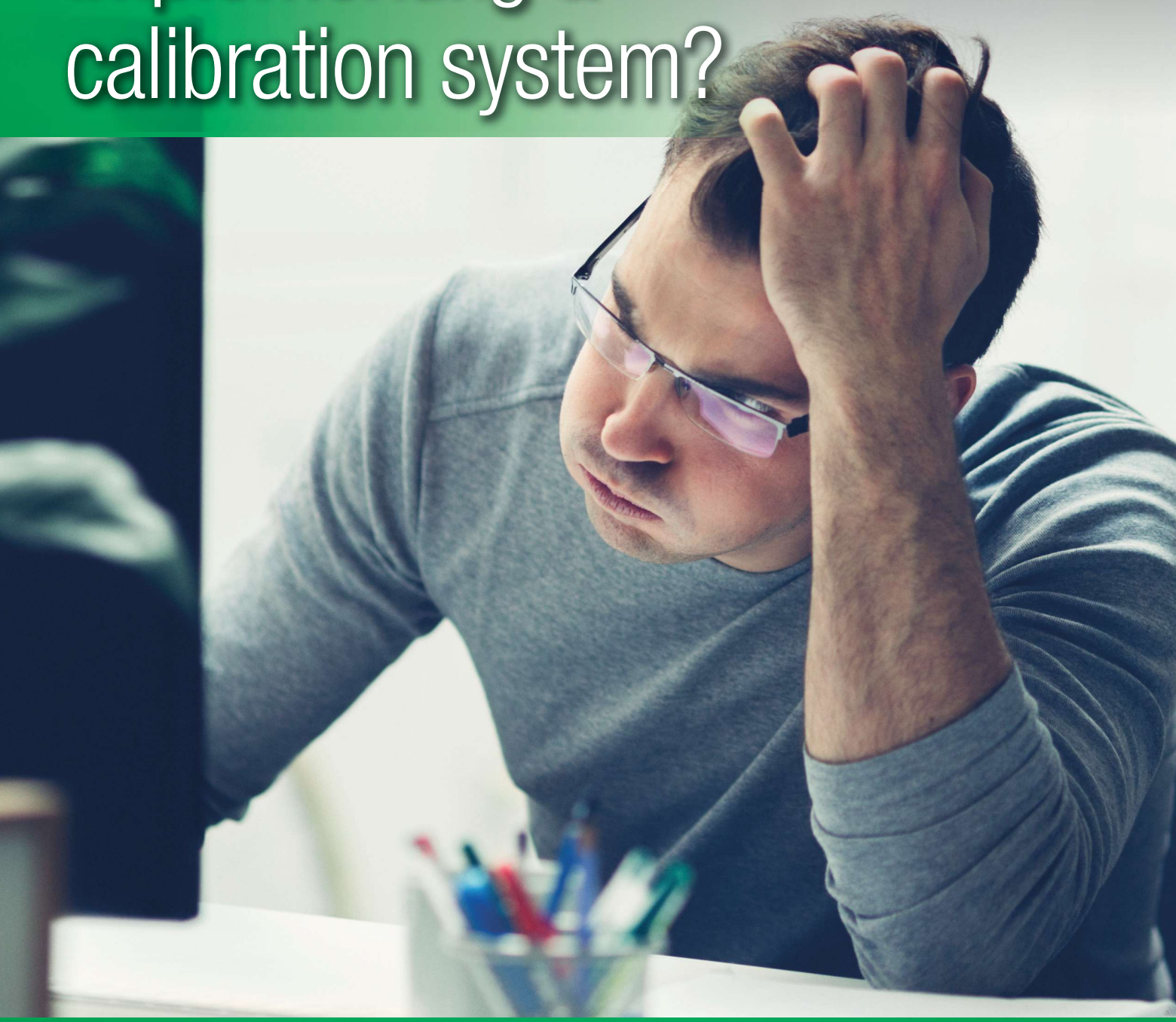
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Automation change agents



By Bill Lydon, *InTech*, Chief Editor

To create the greatest value for their employers, automation professionals need to be change agents. In my experience a *change agent* is a person with specialized knowledge and know-how who facilitates positive change to improve operations efficiency.

Industry today is in great need of automation change agents skilled and knowledgeable in a wide range of technologies. Today there are many new technologies that can be applied to improving production. These choices can easily overwhelm nontechnical management in a company to a point where they simply shut off new ideas and maintain the status quo. One of the obstacles to change is the idea that, "we are using only a fraction of what we have today, so let's not invest in new technology." This ignores the fact that hardware and software built with newer technology have inherently greater performance, integrate more easily, and are simpler to use. For example, I doubt that I fully exploited all the features of my Blackberry, but when I moved to a new smartphone I became more productive with easier-to-use features and functions. I would not be as productive today if I stayed with the old technology.

In many companies, other areas such as sales, marketing, accounting, and information systems groups have been better at selling new technology investment ideas to management than industrial automation groups. There is a real need for automation professionals to be change agents to have a positive impact helping an organization understand the possibilities, show how to make them a reality, gain organizational support, and convince management to invest in new technology.

Automation professionals are in a unique position due to their interaction with a wide range of groups in a company. This gives automation professionals a broader perspective to understand more technical possibilities and how they can be integrated into production.

Automation professionals can learn and refine a number of skills to be effective change agents. Change agents have a high tolerance for ambiguity, functioning comfortably and effectively in an uncertain environment. This is familiar territory for automation professionals who face ambiguity routinely when implementing new applications and solving system problems.

Industry today is in great need of automation change agents skilled and knowledgeable in a wide range of technologies.

Change agents need organizational networking skills in establishing and maintaining appropriate contacts within and outside the organization to understand possibilities. In many cases gaining help from others to implement a new change requires negotiation and positive interpersonal skills such as listening, collecting appropriate information, identifying the concerns of others, managing meetings, balancing conflicting goals, and resolving conflicts.

Being able to communicate clearly and effectively to management the need and value of implementing changes is a critical skill automation professionals need to develop to be effective change agents. Communication needs to convey factual, financial, life-cycle return on investment, and emotions including enthusiasm to stimulate others to get involved. Pulling all this together creates a desirable and challenging vision of the future.

Consider developing the skills to be an effective change agent to deliver more value to your employer, and in the process you will become a more valuable employee. ■

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Salary humor

Loved your article on salaries ["Your recipe for maximum salary," September/October 2014 *InTech*], and these points were priceless:

- "Remain an automation professional for the rest of your career. Let's be realistic, what else are you going to do?"
- "Editor's note: results may vary depending on attitude."

Stephen Rader

Defining big data

I read your article on industrial big data ["Industrial big data analytics," July/August 2014 *InTech*]. I want to understand one important point around big data analytics. The IT industry uses three "Vs" to categorize a big data scenario: volume, variety, and velocity. But, more often than not, I see *V* for volume as a predominant factor. For example, if I was tracking a fleet of vehicles (vehicle OBD, etc.), then the volume will be quite high. But, if I were able to take a photo of the exhaust from the tail pipe of my car and upload it, then I am adding variety too. So, I wanted to know, how does industrial big data analytics cover the variety of data?

Nagesh Subrahmanyam

Editor's response:

I've attended many presentations on big data and analytics; most professionals recommend the variety of data should be defined by subject-matter experts based on the application. Another view is you should capture and store every bit of sensor data that is instrumented, because later you may have a use for it.

Transitioning processes

Nice article on the Internet of Things ["Internet of things: Industrial automation industry exploring and implementing IoT," March/April 2014 *InTech*]. I love and totally agree with the point you make with respect to the manufacturing processes requiring huge transitions and not just the automation technologies. I am not sure all the folks I have talked with realize this. It is a very important point. Well done!

Peter Martin





Source: Automation.com



TRUE STORY

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Honeywell opened an interactive demonstration center in Houston to give its industrial customers a glimpse into the future of managing manufacturing operations. The Customer Experience Center (CEC) brings to life many high-tech innovations for the oil and gas, refining, petrochemicals, mining, power generation, pulp and paper, and other process industries. The center includes, for example, a plant control room outfitted with the Experion Process Knowledge System Orion Console. The console is a completely new interface that integrates larger screens, touchscreen displays, and mobile device capabilities to assist plant operators.

"There's only so much someone can learn about a new technology by talking about it. The goal of the new Customer Experience Center is to physically show our customers what these systems look like and how they can help them solve their most daunting manufacturing challenges," said Olivier Biebuyck, global vice president of marketing and strategy, Honeywell Process Solutions. "Even seeing the setup in a smaller scale like this gives our customers a much better idea for how the technology works to improve the safety, reliability, and efficiency of their plants."

Other CEC highlights include field instrument technologies, such as transmitters for monitoring pipelines, that have been transformed in recent years into smart devices that deliver more relevant information to operators in the control room and to mobile operators. The center also demonstrates how safety, physical, and cybersecurity systems can be integrated in facilities. ■

ISA-100 wireless standard gains IEC approval

ANSI/ISA-100.11a-2011, "Wireless Systems for Industrial Automation: Process Control and Related Applications," has been unanimously approved by the International Electrotechnical Commission (IEC) as an international standard. It will be published by year end with the designation IEC 62734. Since its initial approval by the American National Standards Institute (ANSI) in 2011, ISA-100.11a-compliant devices have found wide global use. There were more than 130,000 connected devices reported in 2012 and more than 1 billion hours of operational service at customer sites.

ANSI/ISA-100.11a was originally developed with international collaboration following ISA's open consensus process as accredited by ANSI, which requires participation and voting by experts from multiple stakeholder groups including end users and suppliers—ensuring that all views and needs are taken into account. ISA100 voting members, including those from end-user companies deploying wireless systems in industrial applications, overwhelmingly voted to approve ISA-100.11a.

ISA-100.11a/IEC 62734 provides reliable and secure wireless operation for monitoring, alerting, supervisory control, open loop control, and closed loop control applications. The standard defines the protocol suite, system management, gateways, and security specifications for wireless connectivity with devices supporting limited power consumption requirements. The focus is to address the performance needs of process manufacturing applications, which include monitoring and process control where latencies on the order of 100 ms can be tolerated, with optional behavior for shorter latencies. ■

ABB doubles oil and gas center

ABB celebrated the official grand opening of its Measurement Products business unit expansion in Bartlesville, Okla. The facility designs, develops, and manufactures natural gas and liquid measurement products and analyzers, as well as software applications and system solutions for the global oil and gas market. The project doubles the facility size by adding 100,000 square feet of space and is expected to double the size of the local workforce over the next 10 years by adding 250 jobs—a majority of which are expected to be in technical and engineering disciplines.

The $14 million expansion is part of ABB's ongoing investment in its North America oil and gas operations and will help meet regional and global customer demand for its measurement products and engineering and project services. This is ABB's third expansion of the Bartlesville site since 2000.

"The strength of the oil and gas market, specifically unconventional drilling activity, has provided our business a great opportunity for growth in recent years," said Gayle Lester, general manager of the Bartlesville Measurement Products facility. "The Bartlesville community offers a talented, high-quality workforce that has helped us to succeed and grow over the years, and we are very pleased to be again expanding our operations here."

Located about 50 miles north of Tulsa, the Bartlesville facility is focused on producing measurement and automation solutions for the upstream oil and gas market segment. These solutions are electronic flow measurement devices for custody transfer applications, remote terminal units (RTUs), software to optimize well production, gas chromatographs that analyze composition and measure gas quality, tank level measurement devices, as well as pressure and multivariable transmitters. ■

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ISA106 and the importance of automating manual procedures

Lessons learned in batch process automation can make continuous processes run better during critical transition times

By Bill Wray, P.E.

I have many years of experience in process control engineering, starting with continuous processes and later with automating batch processes. I find batch automation to be largely about sequential operations and states. To a lesser extent, sequential operations and changing states exist in many continuous processes as well.

Extending the lessons learned in automating batch processes to continuous processes seemed like a natural progression, and I recall discussing this concept with colleagues at technical conferences. In 1999, I got the opportunity to put this idea into practice. When a new technology was commercialized, we converted a batch reactor system at our plant to a continuous process. The operations staff had more than 10 years of experience operating highly automated batch processes. They realized the benefit of sequential automation for managing state changes and transitions, so they strongly supported implementing procedure automation for the new continuous process.

Once the decision was made to design for multiple products, as well as campaigning on the new process, procedure automation made sense. The original batch programming was also a strong basis for automating the new continuous process. The plant started up in late 2000 and used procedure automation for startup, shutdown, restart, and rate changes. We added an automated deinventory procedure and a rate optimizer later, as well as other functions and corresponding procedures. Adding a second product manufactured in the process ushered in additional changes.

That experience made me a firm believer in the value of procedure automation, and in the following paragraphs I will share my experiences as a long-time user. I hope to encourage you to implement procedure automation on your process to benefit as we did.

Why procedure automation?

Procedure automation is not new, nor is it some theoretical concept. It has existed for many years, mostly in batch and semibatch processes, but also in continuous processes by some forward-thinking users. It is all about automating specific tasks in a process that typically require a lot of manual intervention from operators, because many problems originate in manual intervention (figure 1).

There are real benefits in its application to continuous processes, and even a casual examination shows all sorts of opportunities. Procedure automation is beneficial not only in chemical and petroleum processing, but in any process with sequential operations. Procedure automation has been applied to a wide range of processes from offshore platform operation to cracking furnaces to processing nuclear materials.

FAST FORWARD

- While most batch processes use automated procedures extensively, most continuous processes still use manual procedures.
- Manual operation has a high potential for problems due to the variability of how individual operators perform procedures.
- ISA88 concepts and success have been leveraged in the development of ISA106.

Procedures exist in all processes. Sometimes they are written down on paper, sometimes kept in digital form in a document control system, and sometimes they are in somebody's head (tribal knowledge). Well-written procedures enable safe, consistent operations. In fact, in some industries, written procedures are a legal requirement. However, having procedures and following them are often different things. To maximize the benefit of any procedure, it must be faithfully followed day in and day out. Thus, the basis for the main benefit of procedure automation: it enforces adherence to the procedure.

Benefits of procedure automation

Automating procedures enforces consistent operation of the process, resulting in improved quality and higher throughput. Operators are able to operate with fewer errors and delays, creating maximum utilization. This translates into real dollars to the bottom line—something

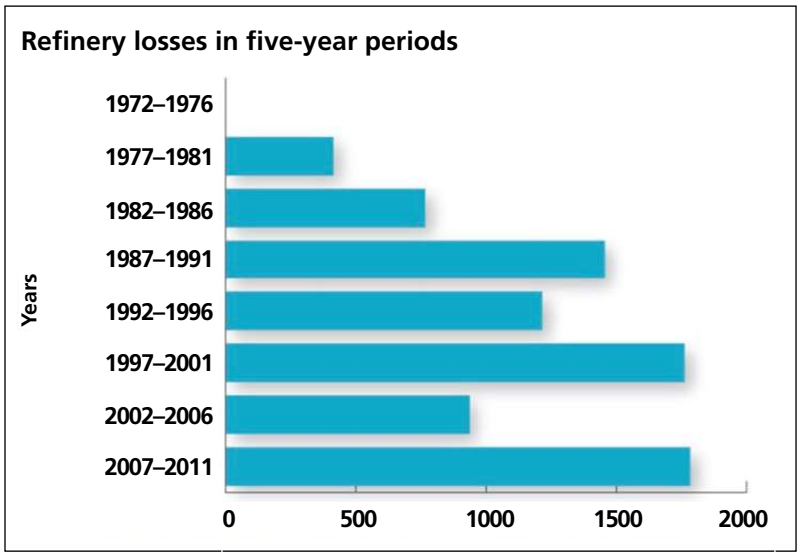


Figure 1. A report issued in 2012 by the Energy Practice of Marsh Ltd., a division of insurer Marsh McLennan, examines the five-year loss rate (adjusted for inflation) from 1972 to 2011 in the refinery industry. Incidents during startups and shutdowns continue to be a significant factor.

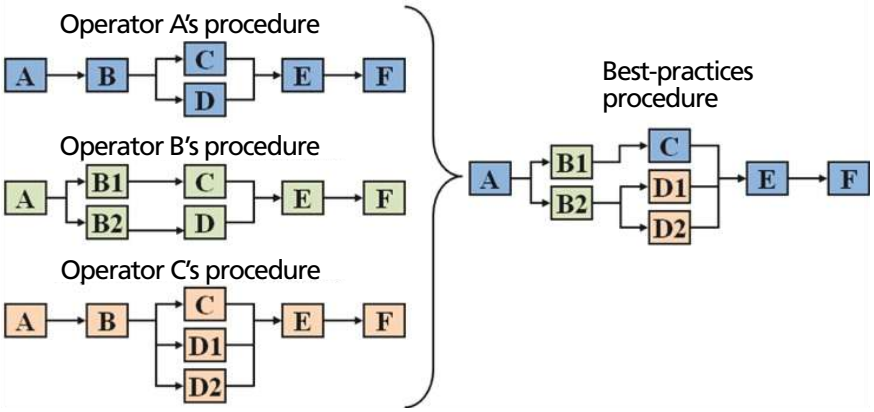


Figure 2. Many procedures are not well documented, and different operators may practice them differently. Part of automating such procedures is determining the best practices and incorporating those approaches, so they will be followed consistently.

any manager can appreciate!

More importantly, procedure automation contributes to safe operations. As previously mentioned, it reduces the opportunity for operator error. In one study, the ASM Consortium found that approximately one-third of incidents were caused by procedures being used incorrectly or not at all. Further, procedures performed less frequently are less familiar to operators and more subject to failure, and these situations generally occur in abnormal operations where the risk of a safety incident (and likely the consequences) is larger. Procedure automation also provides a means for a controlled shutdown, which is less hazardous than an emergency "crash" shutdown.

Procedure automation improves safety through automated management of shutdown and alarm systems in the process. Procedure automation also helps coordination between the basic process control system (BPCS) and the safety instrumented system (SIS), ensuring functions specific to a given operating state are recognized by both. This is especially important during transients or processes where steady-state conditions change due to factors such as different product grades or feedstock. Using procedure automation combined with appropriate SIS programming can eliminate manual startup bypasses of interlocks.

Alarm management is as an important contributor to operator awareness and hence safe operation. Here as well, procedure automation can be used to help manage alarms (e.g., determining which alarms are active or inactive at appropriate times and avoiding "chaff" that confuses operators), especially during severely abnormal conditions, such as an emergency shutdown.

The benefit of improved safety is often difficult to quantify in dollars, but the cost of unsafe operation is painfully obvious. Losses from safety incidents in the U.S. alone are estimated at $10 billion annually. Single events can cost companies upwards of $2 billion, not to mention many lives lost.

Earlier we considered the issue of operator familiarity with procedures and the effects on operators' abilities to properly execute those procedures. As we all know, our workforces are changing, as senior operators with 30 or more years operating experience are retiring and new blood is entering our control rooms. Senior operators that I have worked with generally do a good job mentoring the new hands, but the knowledge transfer is never complete. Procedure automation is a great tool for knowledge retention, and having well-documented procedures and automating those procedures greatly improves personnel transitions (figure 2).

There are other benefits to procedure automation, which have been outlined by the ISA106 committee, and an extensive list can be found in Section 5 of ISA-TR106.00.01-2013 (www.isa.org/store/products/product-detail/?productId=115958).

Implementation step one: planning

As with any engineering project, time spent up front defining the scope will pay benefits later in the project. But there is another detail to cover first, as it is vitally important for all operational areas to be involved in this activity. You must have your operations staff buy in to make the project successful. After all, you will not be running the plant, but they will. Unless they perceive a benefit from procedure automation, it will not be used to the extent it should be. At a minimum, get an experienced operator assigned to the project team.

To succeed as you automate procedures, you must first identify the appropriate procedures for conversion to automation. If well-documented and verified procedures exist for the process, you have a solid foundation. If not, you must create the procedures, and this is no trivial task. It has the benefit of establishing a single correct way to run the plant, however, and often this alone is a huge step forward. Even if well-documented procedures exist, a detailed review to ensure they are actually the correct procedures is recommended.

A team knowledgeable in process operation should review existing procedures for completeness and correctness. The team should solicit ideas for improving the procedures as well. Focus on procedures for non-steady-state operations, as these will be most beneficial for automation. In fact, it may be wise to review existing procedures with a discussion of what areas to automate. The choice of what to automate depends on several factors, including operation frequency, complexity, and the consequences of incorrect execution. These factors will also likely affect the priority of execution.

You might be tempted to embark on a complete automation of the process, from startup to shutdown and all possibilities in between. Although this might be the ultimate goal, your chances of success will be better if you start smaller.

Choose a high-frequency operation of moderate complexity for your first

automation. This combination will offer you the best opportunity to demonstrate the capabilities and benefits of procedure automation. You do not want to bite off too much at first, and it will likely take a few executions to work out the bugs.

On the other hand, you also do not want to solve a trivial problem, as the combination of frequency and complexity will give you ample opportunity for debugging and later demonstrating the usefulness of procedure automation. Another factor to consider is the opportunity to apply the first procedure to multiple operations or operating units.

Another major decision is the degree to automate a given procedure. There is a wide spectrum, from a system that merely instructs the operators at each step and perhaps confirms its completion to another that an operator initiates by pushing the start button and then lets the system execute unaided, reporting back when complete.

Available hardware in the process will affect this decision because some operations likely have components (block valves, pumps) that cannot be remotely activated. In these cases, the automation system will need to guide the operator on what and how to manipulate at the appropriate time.

This may be especially likely on your first project, as management may be initially unconvinced of the value of automation and therefore unwilling to invest in remotely operated block valves, or it may have prejudices against remotely starting pumps.

A related consideration is what degree of operator action will be allowed during execution of the automated procedure. Can an operator override the procedure without disabling it? Can he or she change set points or controller modes? In my experience, the answer to both these questions should be "no," because procedure automation largely relies on conditions being followed as written. However, the operator must be able to take control away from the automation should something go wrong. In this case, be sure the operator gains access to all controllers, so he or she can do what needs to be done.

Implementation step two: design

I typically advocate for a system automated to the highest degree possible within the physical capabilities of the process. If necessary, you can initially trade off overall scope for degree of automation, i.e., fully automating some operations, which can later be incorporated into a larger overall automation system.

If you successfully automate small portions, demonstrating especially to the operations staff the usefulness of procedure automation, you will gain important allies in support of future automation projects. These smaller modules, if properly designed, can be combined into a larger overall procedure. Early victories, even if small, are strategic in the overall effort.

The existing hardware in your plant might also be a factor in your choice. Although automated block valves and pumps can more comprehensively automate your procedure, it may be difficult to justify adding these on your first project. Keep in mind that the lack of these does not preclude automation,



it merely affects the scope.

Whatever degree of automation you choose, it is vitally important that the system informs and involves the operator. The potential problem of unaware operators is more likely with a fully automated system. Messages displayed on the operators' consoles to convey status are useful. A prompt or guide message requiring operator acknowledgement or confirmation is also useful, especially at major transitions, such as the beginning and end of a procedure. Details of implementation depend on the capabilities of your automation system.

Operator awareness can also be improved through training. If the operator knows what to expect, it is easier to follow the automation procedures. If your automation is part of a system replacement or migration project, make sure to train in both the new system as well as your specific application. Timing of the training is also important. Ideally, an operator should be able to put new training into practice as soon as possible after its completion.

As with batch automation, modularity is a powerful tool. If you pull the overall procedure apart into smaller procedure modules, you can develop less complex programming than it would be for the larger operation. Eventually you can combine these into an overall implementation. In this manner, you can build to your ultimate goal of total automation of all operations. You will also find that debugging small modules is much easier than debugging monolithic programs.

Whatever degree of automation you choose, it is vitally important that the system informs and involves the operator.

Depending on your process, modules should be designed to be reusable. Typically, certain design features are repeated throughout the process, and the procedure for operation is also repeated. This can be as simple as starting or stopping a pump, or as complex as operating a cracker or still.

For example, in an olefins plant there are typically multiple cracking furnaces of the same design. Automation written for one furnace and its components can be reused on all furnaces if properly written (e.g., use of generic logic). This consideration also applies for different plants and sites across the enterprise.

Finally, you must consider exception management. What do you do when a valve does not go to its commanded position? What is the response to an unexpected process condition or an operator overriding the automation? How do you detect these exceptions? How do you recover from an exception? Exception management, especially recovery logic, is a major part of any automation project. Procedures can be written for exceptions, but these may be vague and difficult to automate because you simply cannot anticipate everything that may occur during such abnormal states.

You may opt to "soft fail" the automation and turn the process back to the operator—but only if it is made clear that the operator has control. In this case, you must also decide if reentrant logic for the procedure is desirable, and if so you must consider how the automation will resynchronize with the process.

If the exception is critical, a process shutdown may be required, either an SIS-actuated shutdown, or perhaps a more controlled shutdown. Again, you need to consider synchronization of the SIS and BPCS. The correct answer to these questions depends upon your process and automation capabilities.

Implementation step three: startup

Debugging your procedure automation programming is certainly necessary. While you can accomplish much of this offline, such as during a factory acceptance test, debugging in the field during

A supplier's view of ISA106

By Dave Emerson

A significant group of process control system suppliers has been deeply involved in creating ISA-106 technical reports and the standard for many reasons. One of the major discussion points of the "great shift change," as older engineers are being replaced by younger workers, is the idea of capturing the knowledge of those experienced engineers and operators before they leave a facility. This has been widely discussed, but preserving the tribal knowledge of automation system and process operations is truly a challenge.

Procedure automation standards facilitate this process. They provide a platform for building tribal knowledge into automated procedures. This is especially important for dealing with procedures such as those involved with plant startup, shutdown, product grade change, and similar operations, since many studies have shown that plants are particularly vulnerable to safety incidents caused by inexperienced operators performing unfamiliar manual functions during those times.

The ISA-106 standard will provide a framework to build correct procedures in the automation system based on accumulated operator knowledge. Careful automation can make such procedures much simpler, more consistent, and safer—avoiding problems resulting from inexperienced operators making poor choices.

There is still much to do in writing all the parts of the ISA-106 series, so we welcome your participation. The committee co-chairs are Yahya Nazer, Ph.D., and Bill Wray, and my colleague and ISA Fellow Maurice Wilkins, Ph.D., is managing director. Wilkins is vice president over Yokogawa's Global Strategic Technology Marketing Center. Come join this critical work and let your voice be heard in the process.

Dave Emerson is the ISA106 committee editor and director of Yokogawa's U.S. Development Center.

Reasons to use procedure automation

- When procedures are performed manually, there is usually large variability in how they are performed.
- Tribal knowledge among experienced operators often makes up for inadequate documentation, but this is being lost to demographic changes.
- Procedure automation is a key mechanism for capturing and preserving operator knowledge.
- The process of collecting and compiling operator knowledge is an excellent opportunity to analyze and institute sound but undocumented procedures.
- Automated procedures, functioning correctly, can ensure more consistent and reliable plant performance with fewer upsets and safety incidents.

startup will be required as well. Using a high-integrity process simulation during testing can minimize debugging in the field, but it will not eliminate it entirely. In many cases your process will not care what the simulation says and will have its own ideas. Modular and reusable automation components can reduce debugging time and should be used whenever possible.

Because operating state changes occur less frequently in a continuous process than in a batch process, there may be fewer opportunities to execute and debug some of the automation. This factor can cause the complete commissioning of the automation to take longer than some might expect. Be sure to communicate this clearly to management so nobody expects miracles on the first day.

Debugging activities also include coordinating and synchronizing interactions among various automation system components, such as your BPCS and SIS. This is where timing issues can raise their ugly heads, and you should be prepared to solve them.

Lastly, plan for additional operator training during startup, as no training is more effective than that done on the live system. This is also a good time to remind the operators of points they covered in formal training completed earlier.

Resources

Most major control system vendors are participating in the ISA106 standards committee to varying degrees, as are some system integrators familiar with the concepts of procedure automation. Participants identified as voting members are the most active and aware of the techniques. Some also have experience applying the technology.

Of course, the ultimate resource is the ISA106 committee output. The first technical report, ISA-TR106.00.01-2013, is available from ISA at www.isa.org/store/products/product-detail/?productId=115958. It covers models and terminology and introduces procedure automation concepts. The committee is currently drafting a second technical report on work processes.

Committee membership is open to all interested parties, so you can participate in the standard development. This has two major benefits: first, you can participate in discussions with many acknowledged experts in the field, even if only as a listener; and second, you will have the opportunity to review and comment on the committee's work before publication.

To join the committee as an information member, contact ISA (www.isa.org/isa106) and request membership, or email crobinson@isa.org. ISA membership is not required to participate on an ISA standards committee.

The value of procedure automation is well proven. Properly designed and programmed, it will improve the repeatability, utilization, and safety of your process. My 14 years of experience operating a continuous process controlled by procedure automation validates this. I am a believer, because I have seen firsthand what procedure automation can do, and I believe you will experience similar benefits with your processes. ■

ABOUT THE AUTHOR

Bill Wray, P.E., began his career as a process engineer, later discovering his passion for process control, a field he entered in the early 1980s. Beginning with automation of large continuous petrochemical processes, he later moved to batch automation and is recognized among his peers for his expertise. Wray was a founding member of World Batch Forum (WBF) and served in various leadership roles. Since the merger of WBF and MESA, he has joined the international board of MESA. He was the first chairman of the ISA95 committee, leading this effort to publication of the ISA-95 Part 1 standard, and is currently co-chair of ISA106. Wray is employed by Bayer Material Science as a senior engineering consultant and holds a B.S. in chemical engineering from Virginia Polytechnic Institute.

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RESOURCES

ISA106, Procedure Automation for Continuous Process Operations
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Automating Procedures in Continuous Process Applications using the concepts of ISA-106
www.isa.org/store/products/product-detail/?productId=122785

"Next frontier: Operator-automation relationship"
www.isa.org/standards-publications/isa-publications/intech-magazine/2013/december/next-frontier-operator-automation-relationship

ANSI/ISA-88.00.01-2010, Batch Control Part 1: Models and Terminology
www.isa.org/store/products/product-detail/?productId=116649



Taking denitrification to the next level

An upgrade of proven technology with 21st century I&C

By Jaime A. Alba, Peter Loomis, Robert Litzinger, Bruce P. Stevens, and Paul A. Miller

In 2006, one of the largest water reclamation facilities in northern Virginia needed to expand the facility from 18 to 24 million gallons per day (mgd) to support future growth in Prince William County, Va. At the same time, new regulations necessitated an upgrade to improve the nutrient removal capabilities of the plant. The new waste load allocations for total nitrogen (TN) were based on permitted discharge flows on 31 December 2010 with a 3 mg/L TN concentration. The Prince William County Service Authority (PWCSA) recognized the need to simultaneously increase flow and nutrient removal capabilities. At the same time, they needed to replace their plant-wide data acquisition and control system (DACS) with a new modern supervisory control and data acquisition (SCADA) system.

This design-build project enhanced nutrient removal and increased capacity, doubling the existing aeration basin volume and reconfiguring to allow operation in either four-stage Bardenpho or modified Ludzack–Ettinger (MLE) modes. It implemented 14 new deep bed denitrification filters for a total of 24, and methanol feed to the filters was automated to be nitrate load paced and controlled by a proprietary software calculation algorithm. Furthermore, PWCSA installed an additional online analyzer for controlling the methanol feed to the filters for redundancy.

The proprietary software calculation algorithm for control is feedforward/feedback based upon flow and influent and effluent nitrate concentrations. This enhanced the operation and reliability of the process and also reduced the risk of methanol overdose by more closely matching the methanol feed to the actual demand. Consistent methanol dose control is challenging when trying to meet low effluent TN and simultaneously maintain a low effluent carbonaceous biochemical oxygen demand (CBOD).

This plant is currently in full operation and in compliance with the effluent requirements.

Process enhancements

Before the expansion, the plant operated with 10 denitrification filters that had insufficient surface area to process the full plant capacity of 18 mgd in denitrification mode. When the filters were operated in the denitrifying mode, flows beyond 12 mgd bypassed the filters to prevent hydraulic overloading. Even though the filters were capable of hydraulically passing the full plant flow, denitrification could not be achieved at higher flows. Based on the processing limitations and operational cost savings, the filters were often operated seasonally, with methanol added only during the winter for the additional denitrification needed to meet effluent TN requirements. During

summer, the plant had sufficient denitrification capability in the secondary treatment (aeration basins) to meet effluent TN requirements, and the filters operated in a polishing mode without methanol addition to remove suspended solids.

During the design phase, the MLE and four-stage Bardenpho processes were selected for implementation based on a wide range of criteria, including capital cost, overall cost, net present value, land requirements, effluent quality, operability, maintainability, and schedule.

The plant was also required to reduce effluent total phosphorus to 0.18 mg/L. Phosphoric acid addition capability is provided in the filter's area if the filters become phosphorus limited. It was anticipated that any phosphorus allowed to bleed through to the filters or added to the secondary effluent will be removed by the denitrification filters and permit limits will not be exceeded. However, provisions were provided for future implementation of phosphoric acid feed.

The number of denitrification filters was increased from 10 to 24 to meet the new projected demands and to be able to handle when filters are offline for backwashing, bumping, or maintenance/repair.

Plantwide control system replacement

As part of this design-build project, the existing plantwide DACS was replaced. It was obsolete, with key components of the system no longer available from the manufacturer.

The facility transitioned the existing DACS to a modern SCADA system as part of the overall implementation, including designing a system with both new process area control panels and upgraded existing control panels. The final system has about 5,000 I/O points, 25 programmable logic controllers (PLCs) with a self-healing fiber-optic ring, an object-oriented human-machine interface (HMI) system, and a historian interfaced with reporting software that integrates the SCADA and laboratory databases.

The engineer of record managed construction and did quality assurance/quality control (QA/QC) for the new SCADA system and the field instrumentation portion of the project. Some of the activities included QA/QC for the new instruments, startup coordination between PWCSA and the subcontractor, onsite response to design/implementation questions and clarifications, development of maintenance of plant operations (MOPO) plans for transitioning existing and in-service systems to the new SCADA system (with the objective of minimizing the effect on plant operations), and developing and continually updating the SCADA project schedule.



New denitrification filters

Furthermore, toward the end of the project, the engineer of record also guided, witnessed, and approved the testing procedures and results for the SCADA system as a whole. This activity included network testing, uninterruptible power supply (UPS) testing, software testing, and PLC programming testing. The subcontractor performed loop testing (operational readiness test) and the functional demonstration test with coordination from PWCSA, witnessed and approved by the engineer of record.

Automation of the methanol feed system

As part of the new denitrification filters implementation, methanol feed to the filters was automated to be controlled by TETRAPace calculation algorithms. The primary method of operation was feedforward/feedback based upon flow and influent and effluent nitrate concentrations.

The calculation algorithm needs the influent and effluent nitrate concentrations. The existing ChemScan unit was used for this measurement. The ChemScan system (analyzer) is a multiparameter, multisample line analyzer that PWCSA had already been using for some time. The plant staff is used to working with it, keeping the system accurate and responsive. However, during the plant expansion, the facility decided to have a second ChemScan unit installed for redundancy and to eliminate the risk of methanol overdose if one of the units is unavailable. If only one unit

FAST FORWARD

- Recent regulations require more stringent limits for the total nitrogen (TN) in the discharge of wastewater treatment plants.
- Consistent methanol dose control is challenging when trying to meet low effluent TN and simultaneously maintain low effluent carbonaceous biochemical oxygen demand.
- A well thought-out process design requires the right control strategies and level of automation and the right instrumentation for process and economic benefits to the plant.

was installed and was unavailable, typical programming utilizes the last known readings; however, this could result in significant over- or underdosing of methanol if the characteristics of the wastewater change, potentially resulting in permit violations for CBOD or TN. Also, overdosing methanol to the denitrification filters could occur due to the variations in the secondary effluent nitrate levels while the analyzer unit is out of service. This increase of the effluent CBOD level may cause a permit violation.

Instrumentation

In designing critical process systems, redundant systems were used. When permit limits of stringent levels are implemented, operations must use the instruments to optimize the facility and then rely on the electronic components to work reliably over time to enable compliance.

The analyzer is essentially a lab spectrometer that is automated to read more than 254 wavelengths through a process sample from before and after the denitrification filters common inlet and effluent troughs. The sample pumps constantly deliver a small stream to the analyzer, and the system alternates between each sample line in a continuous sequence to monitor ammonia, nitrate, nitrite, and ortho-phosphorous. The system also self-calibrates with a zero standard of deionized water. It self-cleans with an acidic solution to keep the total suspended solids and hardness from fouling the flow through the cell and other components in the plumbing manifold. Detection limits and process trends need to be understood, and plant staff must vet data excursions regularly to ensure the expectations of the operators are met with confidence. The data presented in figure 1 shows a nitrite excursion during startup that indicates an upset in the process, which allows the operators to react to the situation, adjust upstream processes, and adjust chemical feed systems if necessary. Figure 1 also displays how the data from the ChemScan analyzers and the grab samples are very close to each other, confirming the calibration of the instrumentation.



Redundant ChemScan

Online analyzers mainly provide intermittent data, which gives the effect of sharp high and low points in graphical terms, so operators must understand that these peaks and valleys are much smoother and time depressed than seen here. Process control decisions cannot be made on single points

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of data that stand uncorroborated. Patterns with repeatable trends and occasional outliers in the data are used to make rational decisions with and after vetting equipment.

Online monitoring devices can help run a facility better, but they are not foolproof and need regular care and inspection. Maintenance instructions and protection of transient electrical power surges must be done to protect the investment in this data gathering infrastructure.

At the same time, and as part of the design-build expansion project, a backup methanol feed pump was installed so that the loss of methanol feed to the denitrification filters (when running in denitrification mode) is minimized during a failure or scheduled maintenance for the main feed pump. Also, as part of the plantwide DACS replacement, the network backbone was designed with a self-healing fiber-optic ring to minimize the risk of losing the SCADA system and with UPS systems to prevent spikes and brownouts from damaging the critical equipment of the control system.

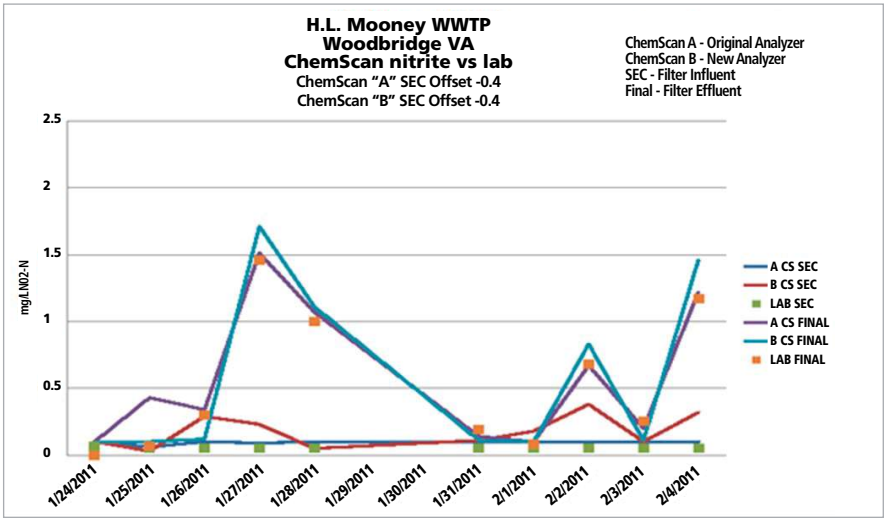
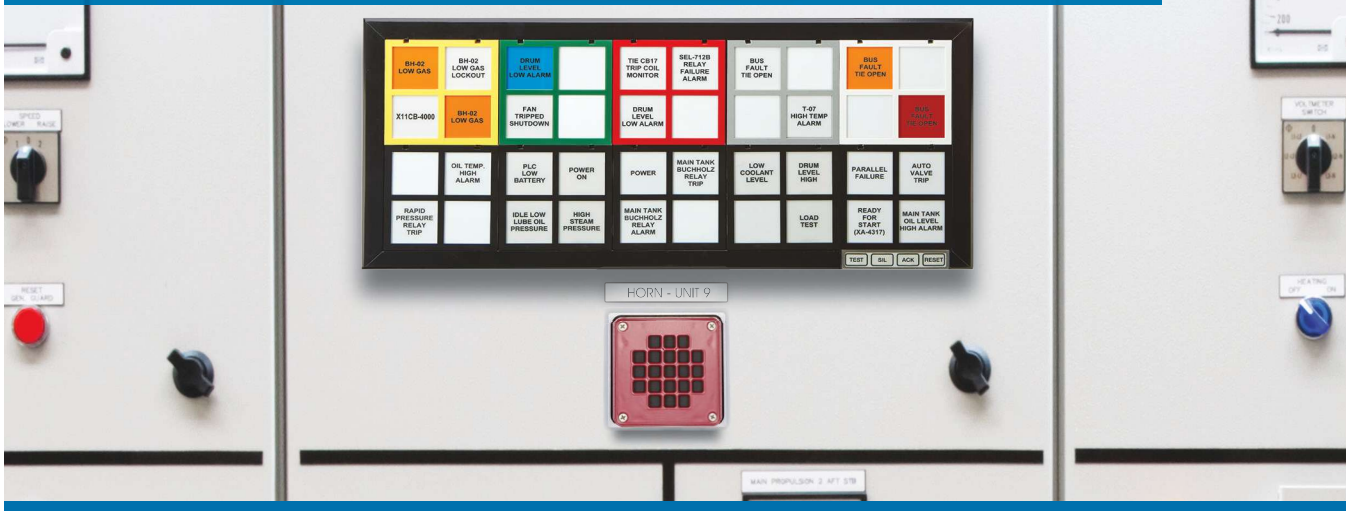


Figure 1. ChemScan nitrite analysis during startup

This whole implementation enhanced the plant operations and reliability of the process, and also reduced the risk of methanol overdose by more closely matching the methanol feed to the actual demand (figure 2). Consistent methanol dose control is challenging when trying to meet

"Our HMI had problems and we didn't have any alarming at that point. But I've never had an annunciator do that to me in 20-something years. I feel more comfortable having that annunciator there."
- *Instrumentation Maintenance Technician, Steelmaking Plant*



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low effluent TN while simultaneously maintaining a low CBOD.

During construction

One of the biggest challenges during this plant upgrade was the coordination, planning, and execution of the MOPO plans. Because the plant had to be kept online during this upgrade, there were many factors to consider to minimize the plant downtime. One of the key factors was to test the hardware and software before it even was shipped to the plant. Once the equipment was onsite, another thorough test was executed to test the field connections and installation.

The startup process was done in phases, taking into consideration that not only the instrumentation, but also the plantwide control system was being transitioned and that there were construction aspects of the project that had to be considered as well. This led to an initial setup and calibration of the TETRAPace calculation algorithms and the analyzer units that later had to be changed to accommodate the different conditions of the phased approach.

Due to the redundant analyzer units setup, the PLC and HMI systems were programmed and configured to accommodate this redundant system. A screen shot of the ultraviolet system area is displayed in figure 3, where the readings of both analyzer units are displayed.

At every operator workstation, which are distributed throughout the plant, the operators have the option to select which analyzer unit readings to use in the calculation algorithms. Figure 4 shows a screen shot of the denitrification filters area where the operators can select which analyzer unit readings to use for the calculation algorithms. This also added flexibility for maintenance of the analyzer units.

Additionally, PLC programming and HMI development for the denitrification filters were implemented as an integral part of the SCADA system instead of having a vendor system communicating to it. This gave the plant staff a more standardized approach to facilitate future changes and maintenance to either of the systems.

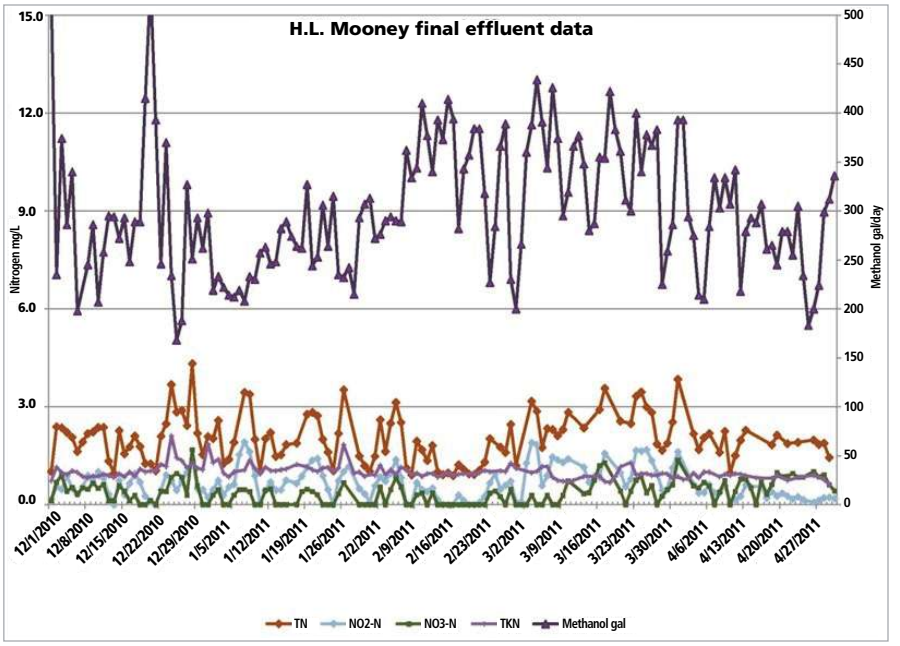


Figure 2. Effluent nitrate and nitrite concentrations, and methanol feed rates after the upgrade

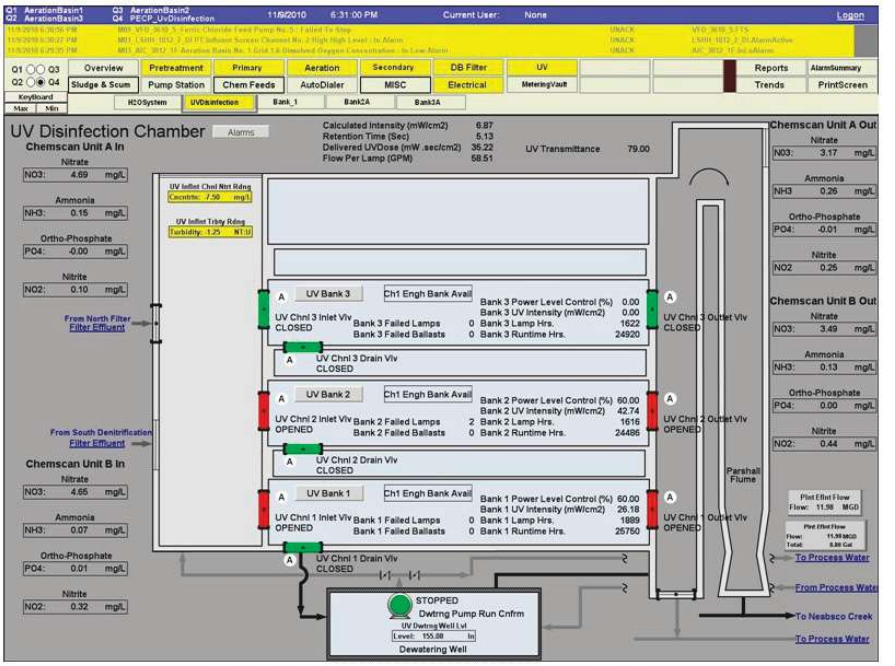


Figure 3. Filters influent and effluent ChemScan units readings

Operational and economic benefits

As already mentioned, one of the main operational benefits is that the risk of methanol overdose was greatly reduced by more closely matching the methanol feed to the actual nitrate and nitrite changes when feeding methanol to the filters. This enhanced control simultaneously maintains a low CBOD. The denitrification filters can now denitrify meeting permit limits up to average day maximum month loading.

At the same time, and as a consequence of the improvements made to the aeration basins, plant operations fed methanol to the denitrification filters until about August 2011, very close to the substantial completion date of the project. Currently, the plant is able to completely denitrify to below



Figure 4. Filters TETRAPace calculation and ChemScan readings selection

3.0 mg/L of TN consistently in the aeration basins. The fact that the plant is currently flowing at about 50 percent capacity might be an important factor for this, but if the flows increase and there is the need for it, methanol can be fed to the filters again.

With the lower flows through the aeration basins, the plant can operate year round in the four-stage Bardenpho mode with maximum anoxic zones. The mode of operation still allows complete nitrification in the aeration basins and maximized volume available for denitrification. At the same time, with the higher retention times, the plant can support a biological phosphorous accumulating organism in the aeration basins, allowing operations to stop all ferric chloride feed to the secondary system while maintaining ferric chloride feed to the primary clarifiers to meet the total phosphorus limit.

On the economic side, the daily average methanol consumption to the filters went down from about 413 gal/day to close to 300 gal/day. Methanol is fed to the filters for an average of 8 months a year, which translates to about 27,000 gallons less methanol used per year. Additionally, since currently sufficient denitrification is taking place in the aeration basins when operating in the four-stage Banderpho mode and no methanol is being fed to the filters, this means that the methanol savings would be about 41,000 gallons per year, above the 300 gal/day of filter methanol that was moved to the aeration basins to improve the nitrate reduction. With a gallon of methanol being about $1.755, this translates into a yearly savings of about $71,955. In other words, what was done before the upgrade to save money on methanol consumption by running the filters in polishing mode has now become normal operations for the plant.

Success

The H. L. Mooney advanced water reclamation facility upgrades became necessary because the average daily flows were reaching 85 to 90 percent of the plant capacity and further growth was expected. Additionally, PWCSA wanted to maintain a waste load allocation based upon 3 mg/L of effluent TN at the future flow capacity of 24 mgd. In addition to the process upgrades to the aeration basins and denitrifying filters, the upgrades included improved controls for the denitrifying filters, allowing a reliable methanol feed control to minimize methanol costs while also ensuring adequate feed.

The modifications to the overall plant have allowed a significant reduction in overall chemical usage for both methanol and ferric chloride. Ferric chloride use was reduced due to the increased volume in the aeration basins, resulting in biological phosphorus removal. Furthermore, the improved controls incorporated into the denitrification filters resulted in a reduction in methanol usage by approximately 25 percent while still meeting effluent nutrient loading criteria. Finally, the methanol reduction at the filters can be traced directly to the use of a load-based methanol addition algorithm for the denitrification filters. ■

ABOUT THE AUTHORS

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View the online version at www.isa.org/intech/ND20141202.

Six considerations for choosing the best-fit motor control technology



Choosing the best fit can help reduce downtime, improve efficiencies, and reduce costs

By Rick Anderson

Industrial engineers are well aware of the electrical consumption burden their motors—especially those running pumps, compressors, and fans—put on their operating budgets. In response, they have turned to efficient motor control technologies that use just enough energy to start motors, provide diagnostics data, and reduce downtime. As motor starter adoption has widened over the years, motor starter technology has also become more sophisticated.

Motor starter methods

Common motor starter technologies include direct-on-line (DOL) starters (or across-the-line starters), soft starters, and variable frequency drives (VFDs). Understanding the application and what is critical for motor control in that application will help determine which starting method to use.

As a basic starting method, a DOL starter applies full voltage, current, and torque immediately to the motor after the start command. A soft starter, or smart motor controller, actively manages voltage to control the starting/stopping current and torque profiles to enhance the electrical and mechanical characteristics of the motor, motor circuit, and machine operation. A VFD converts the AC line voltage to DC voltage, and then inverts it back to a simulated AC voltage for the motor.

DOL methods have the most basic motor-starting capabilities, while VFDs have motor control throughout the start, stop, and run time. For applications requiring motor control only when starting and stopping, soft starters are more economical than VFDs and have more advanced control

than DOL options. In addition, the soft starter and the VFD can start a motor with reduced voltage and current. They promote less mechanical wear, reduce maintenance, and often lead to other operating efficiencies in the system. Although the list of potential application considerations may feel exhaustive, the following questions can help focus decision making:

1. Does the application need speed control once the motor is at speed?
2. Does the application need precise starting and stopping times?
3. Does the application need full torque at zero speed?
4. Does the application need constant torque?
5. What are the cost, size, and thermal considerations?
6. Are there installation and harmonics concerns?

1. Speed control

The first consideration in choosing a motor control technology is the speed control requirements. Some soft starters have limited slow-speed control between starting and stopping. Slow speeds can vary from 1 to 15 percent of the full speed and can be used in a maintenance or alignment operation. Due to silicon-controlled rectifier (SCR) temperature rise and reduced motor cooling, this mode is meant for relatively short-term operation. Once the soft starters transition to full voltage, even though a fixed frequency is applied, the output speed is actually determined by the motor load. The operating speed of the motor cannot be varied, because the soft starter only adjusts the voltage to the motor and not the frequency.

VFDs use a DC bus and insulated-gate bipolar transistor (IGBT) switching to control both voltage and frequency. This allows full and continuous adjustable speed control. If a process requires tight speed regulation, the frequency applied to the motor by the VFD can be changed in relation to the load. In addition, the VFD output to the motor can be any frequency up to the limits of the IGBT inverter or the mechanical limits of the motor.

2. Starting and stopping times

A second consideration is starting and stopping times, and how precise they need to be. Typically, starting and stopping times with soft starters are load dependent. Internal algorithms adjust the voltage based on programmed start and stop times to increase the current and torque to start the motor or decrease them to stop it. If the load is light, the motor might start in less time than the programmed value. If the load is heavy, it may take longer to start. Some newer soft starters have implemented advanced algorithms allowing more accurate and less load-dependent starting and stopping times. On the other hand, VFDs control voltage and frequency, which allows accurate starting and stopping times with minimal load dependency.

3. Full torque at zero speed

For applications requiring full torque at zero speed, a VFD may be the best option. A VFD can produce rated motor torque from zero to rated speed, including full torque at zero speed. Soft starters operate at a fixed frequency (typically 50 to 60 hertz), and full torque is only available at full voltage. Initial torque, or torque available at zero speed, is a programmable value, which usually ranges from zero to 75 percent.

4. Constant torque

Soft starters use voltage to control current and torque. When starting, the current varies directly to the voltage applied, and the motor torque varies as the square of the applied voltage. This means that at different applied voltages, the torque is not always constant. This can be made more complicated with the loading conditions. Some soft starters implement torque control algorithms, but this does not translate directly to constant torque. Conversely, during acceleration, VFDs apply different frequencies to the motor, and voltage is changed in direct proportion to the frequency. This simple VFD control mode is often referred to as constant volts per hertz, and it produces constant torque.

5. Cost, size, and thermal considerations

At low amperage (less than 40 amps), soft starters have a slight cost advantage over VFDs. As amperage and power increase, the cost of VFDs increases more quickly than the cost of soft starters. At high amperages (greater than 100 amps), this cost can be significant.

Regarding size, soft starters have a physical size advantage over VFDs at all amperage ratings due to the physical construction of each device. As current and power increase, this difference can become exponentially large.

In addition, when paired with an electromechanical bypass (internal or external), soft starters tend to be more efficient than VFDs and typically produce less heat. This is also due to the physical construction of the power-switching devices—soft starters have less active components in the circuitry during start, run, and stop modes than VFDs.

6. Installation and harmonics

Installation considerations can be difficult to quantify,

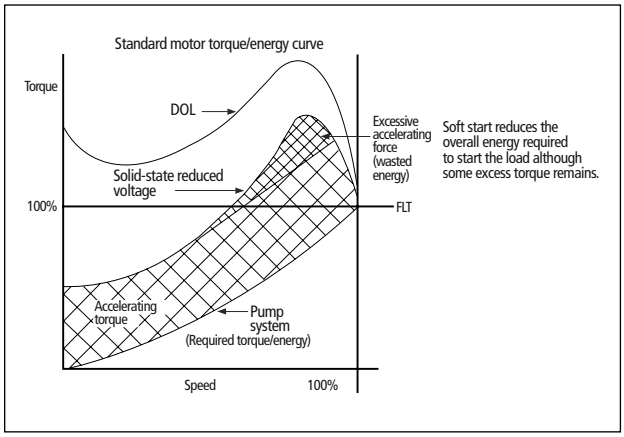
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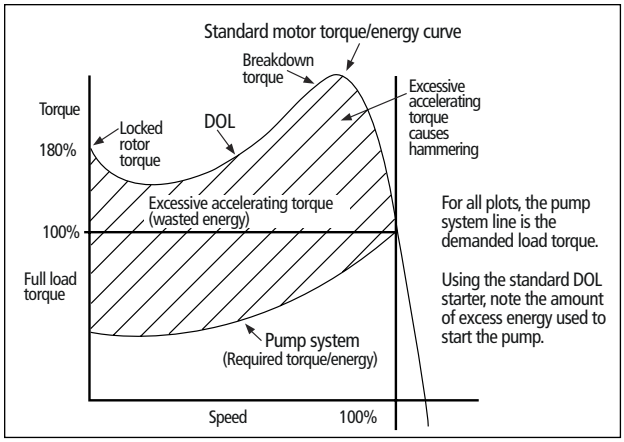
- Engineers have a wide range of motor control options.
- Installation considerations can be classified into a few categories.
- Selecting the best engineering fit reduces downtime and improves efficiencies.

but can be classified into a few rough categories, such as cost, size, temperature, and power quality. Soft-starter installations usually have fewer concerns due to the smaller size and lower costs. If harmonic quality is a concern, soft-starter harmonics are less than those of VFDs, and long cable runs for VFDs need to be considered more carefully than those of soft starters. Special wire types are usually not needed for soft starters, and typically electromagnetic compatibility (EMC) mitigation is not needed.

Assessing soft starters

If, after considering these factors, engineers decide a soft starter is most appropriate, there are a wide array of soft starter types available. Many soft starters have multiple start functions, such as soft start, current limit, pump control, slow speed, and full-voltage starting. For stopping, many perform pump stop, motor braking, and soft stop. In addition, the need for advanced torque and speed control to start centrifugal pumps and high-inertia loads has led to new developments in soft starters.





These new products often have a bevy of control functions and application options, forcing engineers to reassess the performance a soft starter can now provide. Considerations include:

- Advanced starting and stopping methods: In addition to traditional starting methods like soft start, current limit, pump control, and full-voltage starting, there are new exciting control modes.

When to use a soft starter or a VFD

Soft starter

- In lower-to-medium starting torque applications
- In light-to-medium loaded applications
- If little or no speed control is required during run mode
- If reduced mechanical wear and damage to system is required
- If limiting current is the primary reason for not starting at full voltage
- For lower monitoring

VFD

- In single-phase applications on certain drives
- For speed control and system efficiencies operating at reduced speeds during the run mode
- For higher starting torque
- For continuous feedback for critical position control

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  - Linear speed: Regardless of the load type, this method starts the motor in the given time without using an external tachometer. The controller uses just enough energy for both starting and stopping loads, regardless of size.
  - Torque ramp: Engineers can adjust the initial and maximum torque of a defined ramp time for precise control when starting the motor. Soft start mode uses a voltage ramp. In comparison, torque ramp control is much more linear and potentially will cause less stress on the system's mechanics.
  - Slow speed: Slow speed allows short-term operation at speeds other than full speed. Operation at 1 to 15 percent of full speed in forward and reverse without a reversing contactor is useful for maintenance, adjustment, and alignment activities.
- Metering: Metering and operational information is essential for accurately monitoring performance and making timely process adjustments to improve performance and energy efficiency. If users can measure it, more often than not they can optimize it. Built-in energy and power monitoring in soft starters provide real-time energy footprint information for process optimization without additional equipment.
- Energy saver: An energy saver feature uses the soft starter's internal feedback circuitry to monitor the load and recognize when the motor is using less energy—especially common under light loads—and adjust by decreasing the motor terminal voltage.
- Phase rebalance: Uneven line voltages can be equalized using this unique balancing algorithm. The resultant voltage waveforms potentially mitigate the uneven motor winding, heating, and bearing wear that can occur with unbalanced waveforms.
  - Solid-state power structures: Benefits of solid-state power structures include improved performance in high-vibration applications, greater harsh environment immunity, higher operations per hour, scalable thermal ratings, and higher short-circuit current rating.
- Motor and controller diagnostics: User-defined system preventive maintenance, user-defined faults and alarms, tending and fault-triggered data capture, and a real-time clock provide application flexibility, diagnostic capabilities, and improved uptime.
- Network and integration capabilities: Optional communication modules allow soft starters to be integrated into a larger controls environment. This allows improved data flow and plantwide visibility. Networks, such as EtherNet/IP, allow for fast commissioning and easy integration.
- Onboard logic capabilities: Embedded user-defined control technology uses digital and analog inputs and soft-starter internal parameters to control outputs, manage information locally, and perform start/stop functions autonomously. This embedded logic often uses function block programming and can stand alone or be complementary to supervisory control.
- Scalability: Hardware expansion ports accept optional I/O and protection modules. This allows process scalability and optimizations, as well as simple-to-complex application flexibility.

Engineers have many motor control options. Considering exactly what their application requires and choosing the best fit can reduce downtime, improve efficiencies, and reduce costs. The engineering choice now includes new soft starters with multiple starting methods and advanced features. Engineers can match the start/stop profile to the load with a simple control parameter change instead of swapping out control modules. The improved performance and energy savings will lighten the burden motors put on their operating budgets. ■

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Outsourcing:
Defining key relationships



By Paul J. Galeski, P.E., CAP

Successful outsourcing depends on understanding one's own culture, the intended task at hand, and the dynamics of the owner-supplier relationship

Most companies outsource some portion of their project or engineering service needs, because fewer and fewer are fully self-sufficient. Invariably a manufacturer has to decide which elements of the business are critical enough to require internal execution, against those either generic enough to pass to someone else or specialized to a degree that require outside help.

Success in carrying out such a selection depends on understanding one's own internal strengths, weaknesses, and business requirements. Once this is done and agreed upon internally, then the nature of the owner-supplier relationship can be developed to implement optimal engagement.

Some relationships are very straightforward:

- A machine shop may send parts for specialized processing that cannot be done practically or economically in-house, such as electropolishing.
- A chemical company might use a custom packager to meet specialized owner demand, such as delivering acetone in gallon cans rather than drums.

This discussion focuses on a more specialized kind of relationship common to process industries, where the outsourcing company is typically the owner of a plant or facility, and the supplier is a provider of automation solutions, as an example. Although the owner ultimately controls the purse strings and thus has more power than the supplier, the best relationships are those where both parties benefit mutually by giving and receiving the highest value services.

How does a supplier deliver value?

To update the old saying, if you want something done efficiently and effectively, maybe you should not do it yourself. The bulk of any company's effort should be directed toward developing core strategic skills that cannot be effectively purchased outside. Keep in mind:

- Your company cannot be good at everything.
- Assess your internal strengths and weaknesses objectively, even if it means stepping on some toes. Understand what activities truly create a competitive advantage in your industry.
- Concentrate your valuable internal resources on the most strategic tasks related to your continued business and outsource the rest.
- Some functions are better left to experts, just like income taxes. Your needs are best served by hiring people who spend their professional lives working in that specialty.

Engineering is complicated, and relationships with automation solution providers can quickly become very complex. In many respects, increasing the distance between two partners increases complexity and workload, because every detail has to be spelled out and managed. When two organizations have a deeper understanding of each other, projects can become simpler. Everyone knows what to do, and each trusts the other to carry out its part without excessive supervision.

A supplier delivers the most value when a project is carried out in a manner requiring the least management on the owner's end. But this means the supplier has to have a deep enough understanding of the owner to understand just what is wanted. This requires overcoming cultural differences between the process plant owner and the automation solution provider.

Unraveling complex relationships

One of the greatest sources of complexity is a sense of uncertainty when either the provider or owner is moving into unfamiliar territory. When one or the other is not completely certain of what is expected, more will have to be explicitly outlined in an agreement. Some elements will be technical, but others will describe how the two parties are to work together.

Complexity can be minimized if both parties understand the working side of the agreement in addition to the technical requirements. Relatively simple projects should not create an inordinately complex relationship. Some situations are more complex by nature of the work being done, so it is important to understand that multiple approaches are possible and often desirable. Two of the more basic relationships are: hiring engineering services on an hourly basis to augment your in-house staff and using a turnkey control system integration project.

Augmenting your in-house staff by hiring hourly engineering services is a common approach and is suitable for some situations if understood properly. In this approach:

- The project is seen as essentially an in-house effort, but with some extra help needed.
- The company gets the job done with minimal effort invested in building a relationship.
- The supplier's "val-

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- Most every company outsources some aspect of its business, often automation project execution.
- The type of relationship an owner has with its supplier greatly influences the price the owner pays and the quality of service it receives.
- Creating the most advanced kind of relationship is not practical (or even possible) in every situation, but can confer great benefit for large and complex projects or global programs.

ue-add" is low without much investment in the overall success.
- The owner retains all the risk. It is difficult to create supplier accountability, because the owner keeps total control over all aspects of the project.
- There is little opportunity to reduce project risk and little opportunity to increase reward from the relationship.
- The owner "gets what it pays for and nothing more," with no repeatability and no way to leverage the relationship in the future.

For larger-scale projects or those requiring skills not present or available from in-house engineering staff, a turnkey control system integration project with a defined scope of supply is often used.
- This approach can work effectively for both sides if the project is very clearly defined.
- Purchasing departments typically view these as competitive-bid projects in hopes of driving down the initial price.
- Bidders may low-ball to win and then try to ratchet up charges via change orders.
- Such tactics increase total cost of ownership and can create an adversarial relationship.
- The project requires very clear contracts and definitions of who owns what.
- Efficiency and repeatability are typically lost in the bidding process.

Such projects can create a "use the lowest bidder" mentality in a company. Purchasing departments may like it, but it is not always a good approach. If a purchasing agent needed brain surgery, would he or she throw it open to three bidders and pick the lowest-cost provider?

Although hiring services hourly or awarding projects on a turnkey basis are the two most common types of project execution strategies, there are other higher-level methods that can be better for both parties.

Projects drive relationships

Forming more complex relationships depends on the nature of the project and the need and desire to work together for the long term. Deep and longstanding relationships can be built around relatively small projects, and large projects do not necessarily ensure good relationships.

Various terms and acronyms describe these deeper relationships, including engineering procurement contractor (EPC), main automation contractor (MAC), and main enterprise partner (MEP). The definitions are not absolute, and common use causes them to overlap in various contexts. Moreover, the nature of a relationship can evolve from project to project, even when the parties stay the same.

Instead of trying to nail down a specific difference between a MAC and MEP, it is more useful to consider the kinds of relationships two parties can have. Each situation is unique, but much is determined by how the two parties view each other. The spectrum of possible relationships is quite wide, but can be divided into three basic categories:

Episodic but arms length—After working on a project or two, the supplier and owner begin to develop a general sense of mutual comfort. Individuals on both sides get to know their opposite numbers and learn the best methods of contact, what they can expect for response time, typical follow up on questions, and so on. Characteristics of this relationship include:
- Integrator manages the project and provides centralized responsibility.
- Owner may still obtain quotes from other integrators to "keep everyone honest."
- Relationship can last for multiple projects out of a sense of convenience.
- Each project is still considered one-off, and any project may be the last because there is not a strong sense of commitment.

Basic sense of partnership—When a supplier is willing to extend itself for the benefit of the owner without needing to be coaxed, a partnership is forming. From the other side, the owner is willing to accept the supplier's advice rather than insisting on its own way. Overall, trust is beginning to grow:
- The partnership requires work upfront to define needs and expectations.
- An ongoing relationship develops with a mutual sense of strategy for reaching the owner's larger business objectives.
- Services cover all the needs for a project and can meet ongoing expectations after deployment.
- The relationship can reduce total

Owner-supplier relationship levels

Status	Basis	Supplier focus	Supplier spends time	Owner receives	Indicators of success
Approved vendor	Transactional	Learning the organization	Evangelizing, getting the next order	Quotes, proposals	Successful proposals, POs
Preferred supplier	Service based	Gathering input	Explaining what is happening	Custom information and discounts	Timely information
Solutions consultant	Needs based	Business problem answers and expertise	Problem solving	Solutions	Problems get solved
Strategic contributor	Relationship based	Owner's organization	Providing insights	Useful ideas	Repeat business, more projects
Trusted adviser	Trust based	Strategic imperatives that make the owner's business more viable in the long run	Understanding customer needs and delivering appropriate solutions	Safe haven for difficult issues and long-term strategic value	Creative pricing, incentive systems, annuity contracts, high-performing plants

cost of ownership by ensuring a higher level of standards and resource leverage.

Strong sense of partnership—When trust grows, each side wants to make sure the other is satisfied with every project and transaction. The owner receives the service expected, and the outcome of the project is more likely to meet performance, cost, and schedule targets. The integrator is paid fairly for the resources used with a reasonable profit level relative to the value provided. Such relationships do not happen overnight, and the integrator must have resources and infrastructure to deliver on its promises:

- Leadership on both sides must offer full support.
- The integrator must offer a complete suite of services capable of handling all technology needs.
- The ongoing relationship must be cultivated at all levels of both organizations, with shared strategies.
- The integrator must have resource depth to manage the owner's entire technology spend.
- This approach should have the lowest total cost of ownership, but purchasing departments may find it scary at first.

How do you see your suppliers?

Given the above definitions, how do most process plant owners see their automation solution suppliers?

- Approved vendor—A company meets the most basic requirements to support transactions, but it has no specific desirability other than offering usable products and services. If the situation is appropriate, a transaction is permitted, but the supplier is always in a selling mode.
- Preferred supplier—An approved vendor with some desirability outside of its basic product offering, such as special pricing or commercial terms. Other things being equal, this company gets the order, but such relationships can be easily changed when a new supplier comes along with a better offer.
- Solutions consultant—A vendor or integrator offering ideas and solving problems in addition to basic products or services. The supplier has to feel there will be some appreciation for going above and beyond.
- Strategic contributor—At this level, the owner is beginning to depend on the provider as a source of strategic planning assistance. The relationship is moving beyond projects to broader-based challenges.
- Trusted adviser—This is the ultimate level of trust, where the provider and owner are partners, mutually considering the other's position in a relationship. Both endeavor to ensure the long-term viability of the other, as well as mutual value and profitability in any transaction.

The table summarizes these stages of owner-supplier engagement and is a guide for identifying and improving upon existing relationships.

Choose your relationships wisely

Every supplier-owner relationship does not have the potential to advance to the trusted adviser level. Such relationships take time and a great deal of effort on both sides, so com-

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panies have to be selective. It is important to choose carefully and determine when such effort is in order:

- What kind of relationship with this owner or vendor is necessary to ensure the success of this project? Simple projects may only need simple relationships.
- Can you see yourself working with this company again? And again? If you do not see the potential or value a continuing relationship, do not waste your time.
- Do you trust each other? This is one of the most basic but potentially difficult questions, and it has to be answered without qualifications. "Maybe" does not work.
- Does the provider fit your culture? An extension of mutual trust includes shared values at all levels of the two organizations.
- Is this more than an "approved vendor" relationship? The answer may be no, which may be the correct answer. Do not try to convince yourself otherwise if the situation is not appropriate.

Deep owner-supplier relationships require a great deal of mutual effort to establish and maintain, but when they work, they are hugely rewarding for both sides. An integrator is assured of being paid fairly for work delivered without micromanagement, and the owner knows it can expect the best effort possible from the integrator. All projects are delivered with the lowest total cost of ownership, and the owner is assured of ongoing support to ensure gains made are preserved and encouraged. Such relationships are rare, but when they happen, all benefit substantially. ■

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RESOURCES

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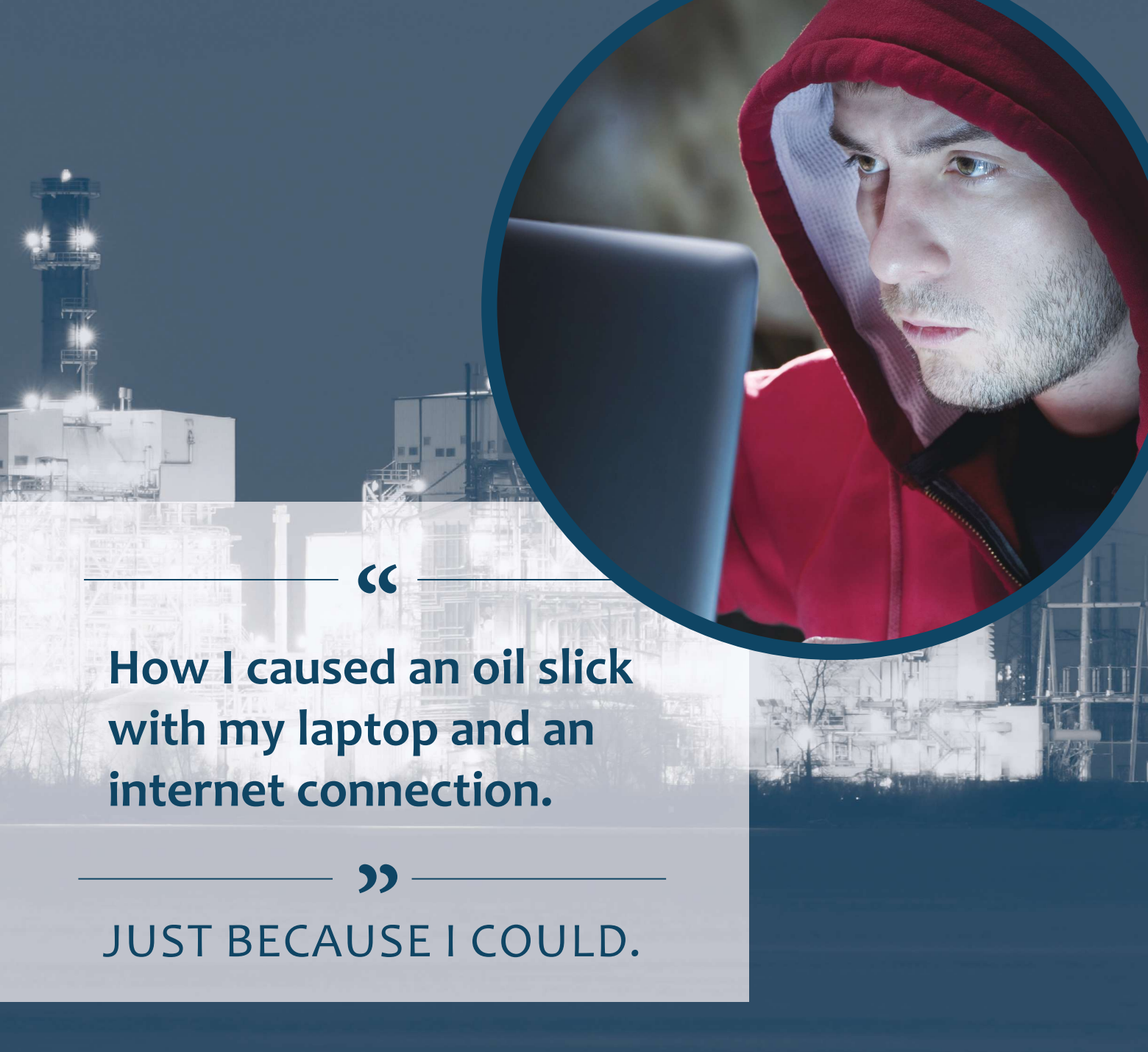
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Drive continuous improvement with information

Report generation is much easier to do than you think

By Roy Kok

These days, lack of data is not the problem, it is the visibility into that data that is challenging. But do not get caught up into thinking that the solution is advanced analytics. There is a step that is far more valuable and easy to accomplish.

We are surrounded by gigabytes and terabytes of data, being dutifully archived by our human-machine interface/supervisory control and data acquisition (HMI/SCADA) systems, historians, and SQL databases. There is comfort in knowing that the data exists should you want to jump in and analyze it. And, no doubt, that is a valuable capability. But if that is your modus operandi, then you will be diving into that sea of data without a suitable context with which to view it. The ability to recognize the unusual in data requires familiarity with the data and systems generating that data. Operators develop that familiarity through their day-to-day interaction with a process. Operators have been known to step into a control room and know through the hum and buzz that the process is running as

it should. That familiarity comes with time.

The second-level metrics are a bit harder to become accustomed to. *Second-level metrics* refers to calculations, based on primary sensor data, which are easily measured and reported. For example, HMI/SCADA calculates and displays secondary metrics for a pump such as run time, off time, cycles per day, peak torque, and motor temperature; it is up to the operator to review these results and become familiar enough with them to recognize a subtle shift or anomalous behavior. We are all well aware of retirement statistics and the workload on operators. An operator's ability to be proactive is being diminished.

Report generation is a solution to this problem—not the report generation of old, but a new class of report generation that makes the selection and presentation of data so easy that you can have any report you want with minimal effort. "Easy" report generation is changing the game. It is likely that you already have lots of experience with report generation. Your experience comes from vendor-specific solutions, designed to meet the requirements of compliance reporting (e.g., Food and Drug Administration, Environmental Protection Agency, state and federal, batch quality, or validation). Compliance reports are the ones you "must have" to accomplish the business you are in. As part of your fundamental business, they will be done at any cost. Hence, the tools do not have to be great; they just have to get the job done. That has been the level of the bar for industrial report generation. As a result, typical report generation goals have been the absolute minimum requirements.

Solutions for report generation also come from bridging business intelligence (BI) solutions, such as SAP's Crystal Reports and Microsoft's SQL Server Reporting Services, into industrial automation applications. This approach typically has several compromises: connectivity to data sources is limited to those with business interfaces (OLE-DB or ODBC); these solutions are not aware of automation-style metrics (OEE, set-point analysis, energy analytics, etc.); ease of use is secondary to functionality—domain experts are typically required for implementation; and a high level of information technology (IT) familiarity is required for solution rollout. These tools are very capable and are excellent for enterprise-level reporting, but BI solutions all have limitations when applied to industrial automation. So, what are the attributes of an "industrial" reporting solution?

Ease of use

"Ease of use" is a feature critical to quality (CTQ) to meet the requirements of industrial reporting. In the world of automation, ease of use also means following the paradigm of the common "tools of the trade." The tools of the trade in automation are configurable and should not be programmer oriented. The eye is on the result, not the process. An engineer wants an HMI, the storage of data, operation of a programmable logic controller (PLC), or the definition of a report. High-level environments delivering this functionality are configurable through menus and selections in a way that any plant personnel can easily learn and apply. For many years, the focus was on the integration of programming environments, such as Microsoft's Visual Studio or VBA, to augment a configurable solution, enabling the "yes" answer to "Can your system do that?" We are in the age of "give me the fish," not "teach me to fish." Again, report generators of the past have never delivered on this "ease of use" CTQ, and so they have limited the opportunity for report generation as a corporate solution for information visibility and continuous improvement. This paradigm has shifted. Purpose-built report generation for the automation industry is now easy and powerful, and report generation is ready to become a significant component of every automation environment.

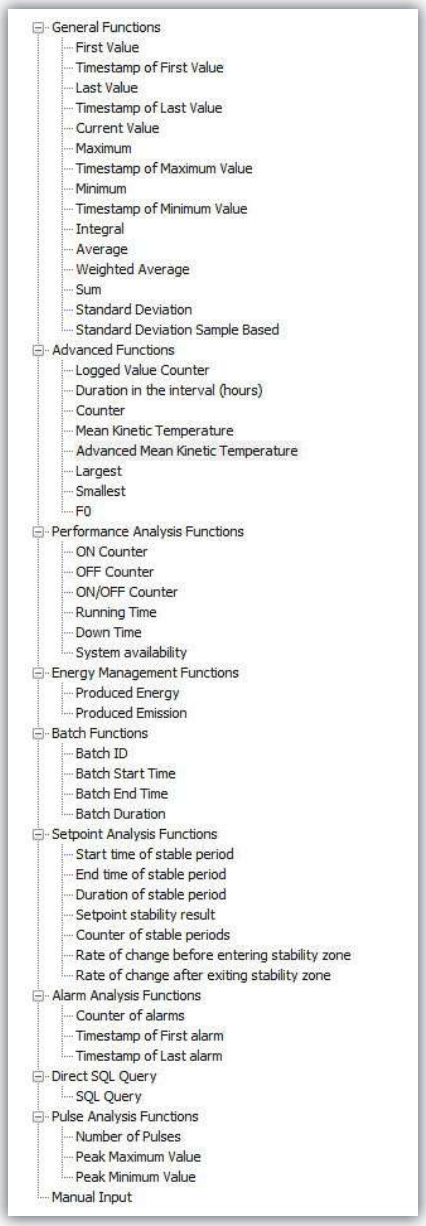
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- The problem is not a lack of data, but visibility into the data.
- The world of automation data needs answers for efficiency.
- IT-oriented business intelligence software comes up short for industrial automation applications.

Connectivity

A report solution for industry must understand industrial data sources. Users benefit most when a report can aggregate and analyze data from a variety of data sources, both business and industrial. From a product perspective, this refers to drivers that can be installed to connect the reporting engine to any source of data that exists inside or outside your plant. Data in enterprise systems, wherever they are, can be queried through business standards, such as OLE-DB or ODBC or by importing CSV or Excel files. BI tools can address only some of these requirements.

Data sources also include industry standards, such as OPC DA, OPC AE, OPC HDA, Modbus, BACnet, and proprietary interfaces to HMI, SCADA, Historian, analyzer, custody transfer, batch, and myriad vendor-specific solutions. Often, these data sources can analyze and return data in advanced ways, and performance is maximized when leveraging this functionality. For example, if you need hourly averages for a day, it is best to ask the source for that statistic rather than request a day of data (potentially tens of thousands of samples) to produce the hourly stats. Of course, if that level of analysis is required, an industrial reporting



Drivers in an industrial reporting solution

solution would package and compress remote requests of data and securely pass that data to the reporting server, performing the task as efficiently as possible.

It is also extremely valuable to recognize the different types of data in industry, for example, real-time, history, and especially alarm data. An industrial report generator understands alarms and provides the ability to intelligently query and display results based on the uniqueness of that data. Ideally, connectivity to disparate data sources will be normalized—presented in a similar way to the user—to remove any data access complexities and allow the user to focus on the task at hand: simple access to the data. BI solutions just do not address this.

Industrial metrics

The world of automation wants answers to automation questions. How many times did that pump cycle? What was the maximum and average temperature reached when the motor was running for more than 1 hour? How long did that batch take? Does that furnace meet the required Thermal Uniformity Standards? Is the building efficiency as expected for today's weather conditions?

The answers can come from any reporting system, but will come "easily" in a system designed for this type of analysis. Ideally, these metrics can be applied to a data query, and the results will be formatted for presentation with little additional effort, because the report generator is smart, and display objects are aware of upstream statistics.

Some process reports are time based, for example, monthly, weekly, or shift reports for production, energy, or quality. However, some processes are event based, displaying irregular start and end times. The latter can be considered analytics on "batches" of data. Examples of the latter are the backwash process of a water filtration system, triggered by a filter pressure and finished at a minimal turbidity level, or any traditional batch process in the food, pharmaceutical, or chemical industries. This type of reporting can be tricky for BI tools.

Batches of data require the identification of the batch and then subsequent queries of sensor data based on the batch start and end time from automation data sources like plant historians. Batches may not be stored ahead of time. Batches may require manual identifications or can be automatically triggered through monitoring a status variable (pressure drop or turbidity) as in our previous example. Industrial report generators can also automatically name batches based on HMI/SCADA tag data or manual selections.

Formatting and presentation

Formatting data to deliver automation information is critical. All report generators offer formatting options for displaying values. But in the world of automation, values need to be displayed in a context, for example in relation to alarm limits or the display of objects that conform to specific standards.

Take, for example, the "Setpoint Analysis Report" on a thermal process and the validation that the thermal processing was a success. This sample report displays a variety of productivity features that are delivered through a set of standard analytic functions and graphic objects that are function aware: the analysis of a set of sensors, the detection of appropriate soak start and end times based on last sensor in and first sensor out, the calculations of duration and rate of changes, the final determination of pass/fail, and the resulting presentation in a "Setpoint Analysis" aware trend chart.

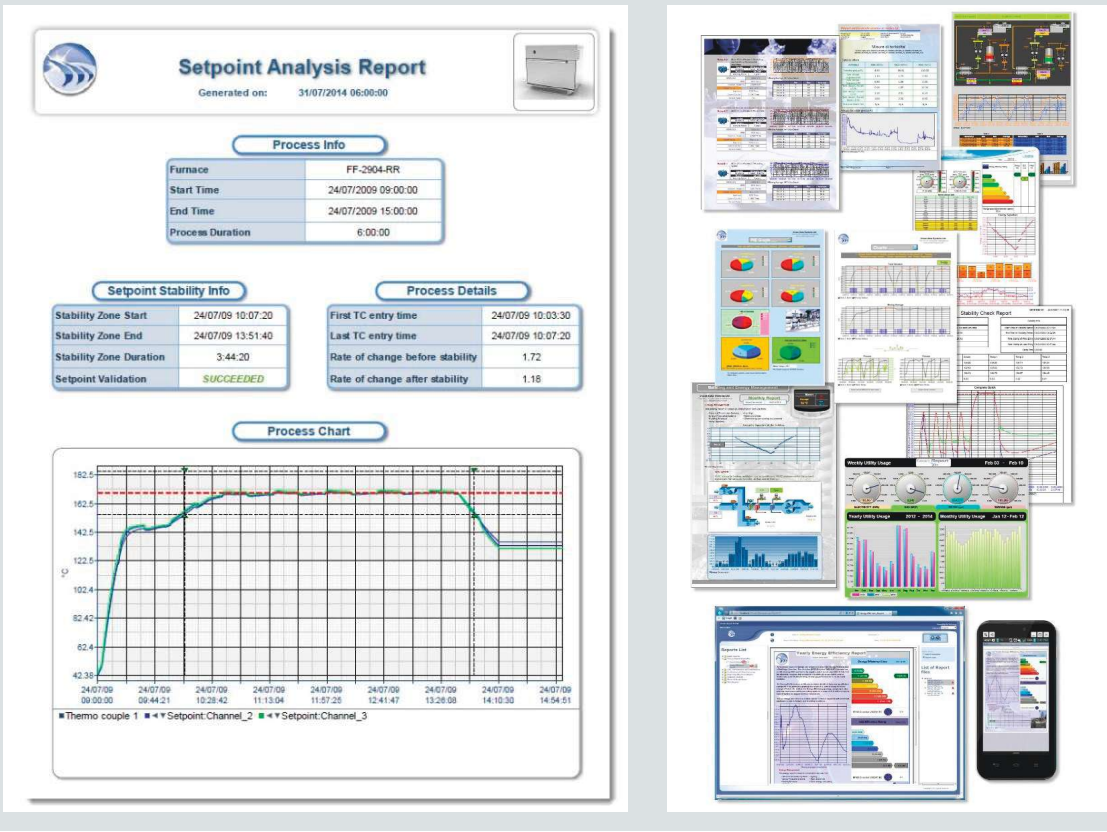
Of course, a reporting solution should also offer a broad range of presentation objects such as pie charts, bar charts, tables, trends, and x-y plots. But in automation, users will benefit from numeric displays, gauges with color bands, graphic images with data overlays, and performance objects.

Manual data entry

Automation data is never complete. There is always the need for some level of manual data entry, either as lab samples or operator observations. A reporting solution for automation has the ability to integrate user-friendly data collection mechanisms. These can take many forms, from automatically importing CSV or Excel data files to data entry forms in a web portal. In either case, the resulting reports will be a combination of automatically acquired data from automation and business systems and information entered manually or through bar codes and other data input methods.

Flexible delivery options

One of the most important aspects of any report generation solution is not the actual generation of the report. It is the storage and delivery of information. Here too, BI tools can come up short. The generation of information spawns the need to intelligently archive said information. This can include automatically naming directories and defining



Sample analysis reports and displays

the longevity of storage, automatically purging that which is no longer needed. After archiving comes delivery. Delivery often includes automatically emailing to individuals or groups. Reports are often delivered to remote archives; FTP should be an option. Finally, reports should be easily accessible through an intranet or Internet web portal. Web portals should support all browser types (e.g., Internet Explorer, Chrome, Firefox, Safari) and both full browser and mobile browser form factors independent of operating system, (e.g., iOS, Android, Windows, or Linux). These delivery options are part of a robust, easy-to-use reporting solution designed for automation systems. Add-on technologies should not be required.

Support for vertical applications

In addition to the features for broad market appeal, report generation needs to address the requirements of vertical applications. The life sciences markets of pharmaceutical and biotechnology require a high degree of security and user authentication, version management, audit trails, and electronic signatures. These features are necessary for 21 CFR Part 11 requirements. Other markets may require high availability and the ability for report generators to support primary and backup printing devices. In even more critical circumstances, reporting solutions may be required to work in tandem, allowing one report server to automatically take over in the case of a primary report server failure.

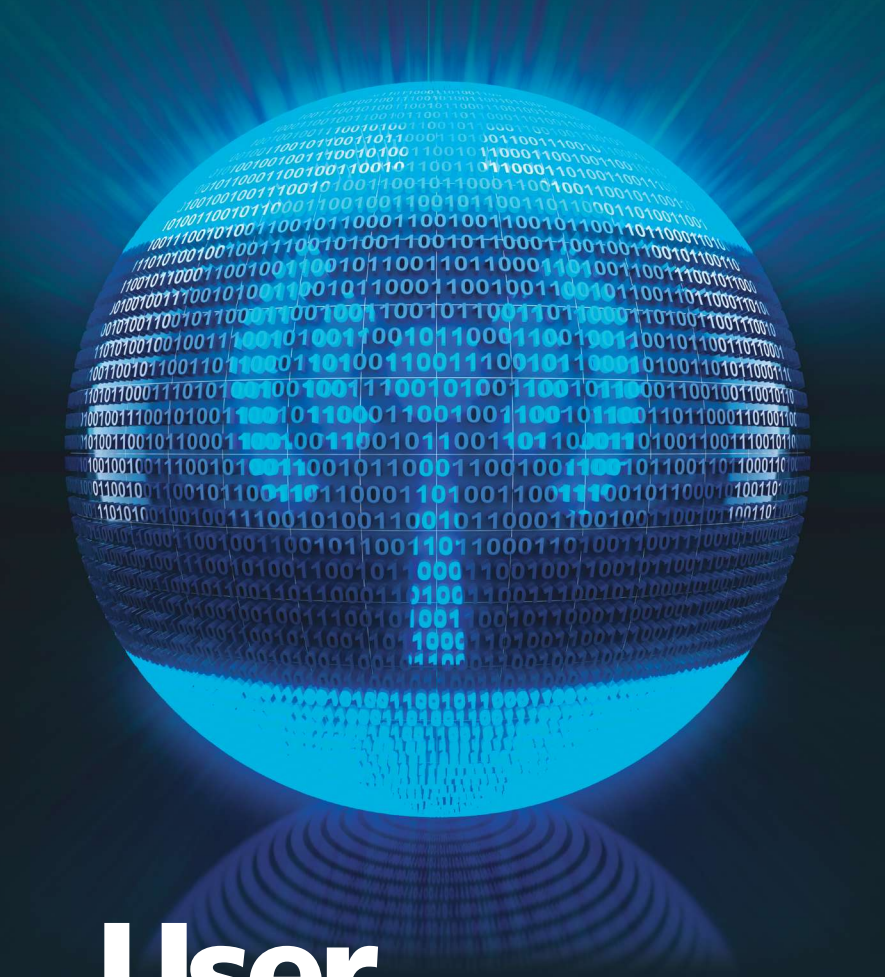
Although BI tools can and do clearly address some of the requirements of the industrial automation marketplace, it is also clear that new purpose-built solutions will offer significant advantages across the board, and especially in vertical markets. Although the purchase price of automation-oriented reporting tools is typically higher, their overall cost of ownership will be greatly reduced due to the savings in development and customization and by leveraging the automation-oriented statistics, easy-to-use configuration environment, and reduced learning curve.

With connectivity to all automation data sources, purpose-built automation-oriented reporting and dashboard solutions can aggregate information from a wider range of sources than BI tools offer. Finally, as a configured solution with all functionality already delivered, these tools display higher reliability and a significantly reduced learning curve, letting you focus on acquiring and delivering information, not getting caught up in IT technologies and system integration. ■

ABOUT THE AUTHOR

Roy Kok (Roy.Kok@DreamReport.com) is an automation industry veteran of more than 30 years, having worked with HMI/SCADA and reporting technologies. Kok is the vice president of sales and marketing for Ocean Data Systems, makers of Dream Report.

View the online version at www.isa.org/intech/ND20141205.



User adoption of industrial wireless

Applications, technology, and systems

By Jay Werb

Industrial wireless instrumentation is rapidly becoming the technology of choice for a growing class of applications. A wireless deployment saves significant costs compared to an equivalent wired installation, resulting in savings of 20 to 30 percent in simple configurations. Cost reductions can be even more compelling in scaled installations or in remote locations. Where wiring is cost prohibitive or infeasible, wireless enables best-practice instrumentation wherever it is needed for efficient and safe industrial operation.

The cost advantages of wireless instrumentation improve with scale. In a wired system, the cost of each additional instrument requires extra wiring and the associated labor, equipment, and maintenance. A wireless system, if designed for scalability, can accommodate additional devices with the same infrastructure and no additional wiring. For the first time, applications with hundreds or thousands of measurement points can be reasonably contemplated.

Until fairly recently, most users and experts viewed wireless instruments as intrinsically inferior to their wired counterparts, with wired instrumentation always being preferred when feasible. As experience with wireless technology grows, this attitude is shifting, with wireless becoming the default user selection for well-proven applications. Today, major users require cost justification for wired instrumentation in applications where wireless has been demonstrated to exceed user requirements.

In Clayton Christensen's model of innovation, adapted in figure 1, low-cost products initially take a beachhead position with simple applications at the lower end of a market. From that starting point, they work their way up-market. A tipping point occurs when mainstream users discover that the low-cost products can be used in high-end applications. Today, most major users are in the "mainstream adoption" zone, which is shown as an oval to indicate that adoption rates vary from user to user.

Wired versus wireless instrumentation

Table 1 summarizes the main differences between wired and wireless instrumentation. Some of the listed characteristics, such as fading and interference, relate to radio considerations. Other characteristics, such as battery replacement, relate to energy constraints when wireless instrumentation operates in locations without power.

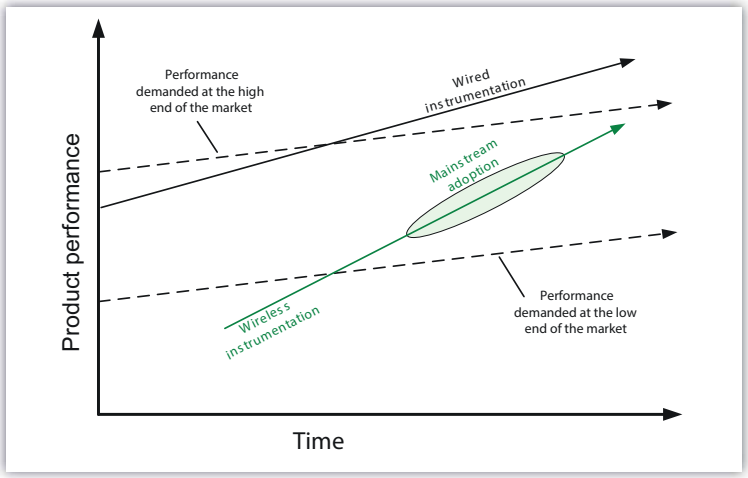


Figure 1. Christensen innovation model adapted for industrial wireless
Source: AIW LLC

FAST FORWARD

- Industrial wireless instrumentation is rapidly becoming the technology of choice for a growing class of applications.
- The cost advantages of wireless instrumentation improve with scale.
- Industrial wireless users are quickly learning wireless performs well for a wide range of applications.

	Wired instrumentation	Wireless instrumentation
Installation	Wiring feasibility and cost • Data and power • Cable length, configuration rules Infrastructure equipment	Access point architecture and placement Range to access points and neighbors Mesh network design
Instrumentation	Full range of available instruments • No battery constraints Proven in use for decades • Generally safety integrity level (SIL) rated Continuous reporting	Partial range of available instruments • Wireless adapters if power available Proven in use for years • Sometimes SIL rated Periodic reporting
Performance	Bus capacity • e.g., 4–20 mA Reliable until the wiring fails • Corrosion, vibration, etc.	Data freshness and availability • e.g., 30 seconds @ 99.99 percent Shared channel capacity • e.g., 90,000 time slots/minute Channel transients • Fading, interference, blockage, etc.
Management	Add wiring and equipment as needed Fix wiring after it fails Monitor instrument reporting	Short-term management: • Redundancy for automatic self-healing Long-term management: • Monitor network diagnostics • Anticipate systematic problems • Reconfigure wireless infrastructure • Battery management • Radio spectrum management
Security	Physical security of device and wire	Physical security of device Management of credentials and keys Network diagnostics
Redundancy	Extra wires for redundancy	Radio mesh Radio as complement to wired link

Table 1. Principal differences between wired and wireless instrumentation
Source: AIW LLC

Major advantages of wireless instrumentation include:

- Lower cost, especially when large numbers of instruments are installed.
- Manageability. When wired connections fail, they are typically complete failures that occur without notice. Wireless failures are usually transient, and those transient problems can mostly be avoided by preventative maintenance linked to wireless diagnostics.
- Flexibility. After a wireless system is installed, it is easy to add new wireless instruments and also to report more data from existing instruments using wireless adapters.
- Security. Wireless security extends to the field instrument and does not rely on physical security of the transmission medium.
- Redundancy within a wireless network. Typically, wired instrumentation relies on a single wire to each instrument, with various opportunities for failure. Field experience demonstrates that a redundant wireless channel can be every bit as reliable as a nonredundant wired channel, particularly when wires are long or subjected to challenging conditions.
- Redundancy at the plant level. A wireless system can add redundancy to wired reporting, with the same data reported through wired and wireless channels. Similarly, when field instrumentation is involved in an independent protection layer (IPL), wireless may be an advantage if another IPL uses available wiring.

Table 1 also suggests a set of generally agreed upon disadvantages of wireless instrumentation at this time. These considerations can be generally grouped as battery-related and radio-related operation.

Disadvantages of battery-powered operation include:

- Battery maintenance. Maintaining the batteries of wireless devices somewhat offsets wireless cost savings. A well-designed wireless solution should ensure that battery replacement occurs in conjunction with an instrument's general maintenance interval.
- Limited wireless instrumentation. An ISA100 Wireless adapter can convert a wide range of existing instruments to wireless, but only if the wired instrument has the power to operate. In locations where there is no source of external DC power, instrumentation options can be limited. New wireless products are being rapidly released to meet market demand.
- Continuous reporting. In some use cases, it is not technically feasible to sample and report process data continuously under battery power. ISA100 Wireless devices can be configured to report process data frequently in critical applications, with predictable battery life impacts, but only if the sensor has the energy to collect the data in the first place.

Disadvantages of radio operation include:

- Procedural barriers. Wired instruments have been used for decades, and processes for specifying and approving wired systems are well established at user sites. Many of these same users do not have clear processes for approving wireless, particularly when safety credit is involved.
- Statistical nature of radios. Radio performance is statistical by nature, with packet errors and retries being fundamental considerations for any wireless system design. A well-designed wireless system will have plenty of built-in margin, as well as extensive network diagnostics that detect loss of margin even while the system achieves its performance objectives.
- Limited reporting rates. ISA100 Wireless is designed to support reporting as frequently as every 0.25 seconds with a transmission latency of 0.10 seconds in structured configurations. Faster reporting rates are considered (by the major standards) unsuitable for battery-powered operation at this time.
- Spectrum management. Wireless instrumentation shares the radio spectrum with other systems and applications. Spectrum management generally needs to be considered when each new wireless system is installed, and should also be continuously monitored. A well-designed wireless system will have performance margins built in and will include extensive diagnostics to detect loss of margin due to radio interference and other considerations. ISA100 Wireless radio diagnostics include metrics that are specifically intended to detect and blacklist problematic radio channels automatically.

Based on real-world experience with all of these factors, users of industrial wireless are learning that wireless can deliver more than adequate performance for a wide range of applications.

Use cases for industrial wireless instrumentation

Industrial wireless instrumentation is being applied to a wide variety of applications today. Figure 2 represents one way to summarize the major application use cases.

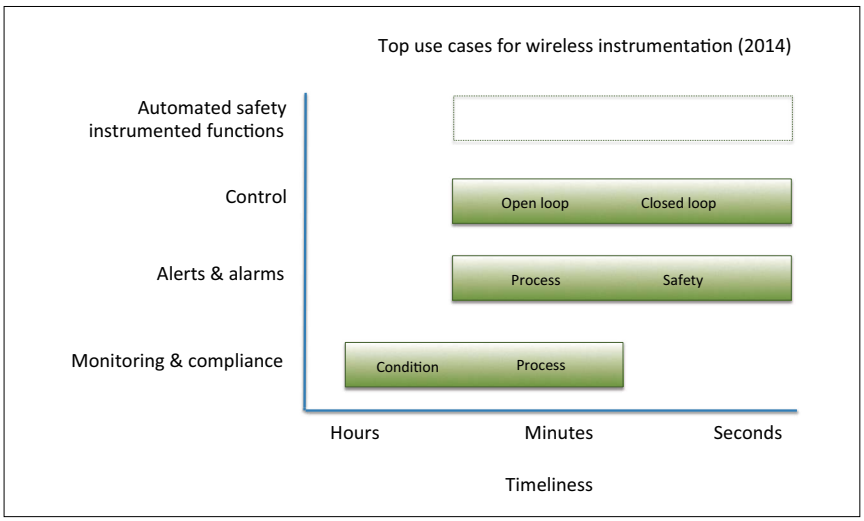


Figure 2. Current landscape of industrial wireless instrumentation Source: AIW LLC

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The vertical axis shows types of applications, derived from ISA100 usage classes:

- Monitoring and compliance applications track the status of equipment or a process state, such as temperature or vibration. Monitoring data is archived for subsequent review and may or may not be displayed to operators.
- Alerts and alarms track the status of a process state, such as temperature, or a safety state, such as hydrocarbon gas level. Exceptions are reported to an operator for appropriate action.
- In control applications, wireless is somehow involved in a control loop. "Open loop" means that a user is in the loop; "closed loop" means that the loop is automated. Closed-loop applications are sometimes divided into outer loop and inner loop.
- In an automated safety instrumented function (SIF), a set of equipment is intended to reduce the risk of a specific hazard in an automated safety loop.

The green boxes in figure 2 indicate where wireless instrumentation is targeted, in actual practice in 2014. This is intended as a statement of fact, not to imply that wireless is unsuitable for other applications. For example, ISA100 Wireless was designed for subsecond reporting rates, but subsecond timing is not shown in figure 2 because few wireless instruments today are so configured. Similarly, wireless instrumentation for automated SIF is shown with a dotted line to suggest that it is feasible but not currently being adopted by many users.

Alerts, alarms, and safety

Alerts and alarms merit special mention due to safety considerations. *Alarming* involves an out-of-bounds condition that is reported to a user or system. Yokogawa Electronics Corporation has reported that more than 50 percent of its wireless instrumentation projects require a 1–10 second update period. Cited applications include gas detection, fire detection, monitoring for operation (power), monitoring for safety (steel), cold temperature monitoring (gas), and tsunami detection. All of these can be considered "alarming" applications.

According to ANSI/ISA-18.2-2009, an alarm is "an audible and/or visible means of indicating to the operator an equipment malfunction, process deviation, or abnormal condition requiring a response." The essential element of this definition is the response to the alarm.

The term *alert* is generally used for applications with less rigorous requirements than alarms, for example maintenance alerts where the user response is not specific or time critical.

A "safety-related alarm" designation can be applied to an alarming application that is a candidate for safety credit as an IPL. To simplify approvals, less than 0.9 availability may be claimed, thereby classifying an alarm as a basic process control system. Regardless of an alarm's classification, high reliability is invariably a key objective.

It is well established that an alarm system must be designed to effectively handle individual alarms during normal operation and handle many alarms during a major plant upset. Wireless has the potential to support many more alarms than has been feasible in the past, so alarm management is an essential consideration in a scaled wireless implementation.

Gas sensing is an important use case for wireless alarming. The requirements vary by application and region. For example, the process safety time of response (t90) for hydrocarbon gas detection as required in IEC60079-29-1 is 60 seconds. In Japan, the time to alarm must be within 30 seconds for a specified concentration.

ISA100 Wireless and the Internet of Things

ISA100 Wireless was designed as an extension of the Internet, building on the IPv6 "Internet of Things" standard. IPv6 is an addressing scheme that allows an essentially unlimited number of devices to be individually addressable. Collections of devices can still be segregated as needed for security and manageability, but with IPv6, addressing considerations are no longer a constraint. Most forecasts suggest that tens of billions of devices will be Internet accessible within a decade. With its use of IPv6, ISA100 Wireless is well positioned to ride that wave.

IPv6 networks use familiar standards-based addressing and can be managed using tools derived from traditional information technology network management tools and systems. Today's

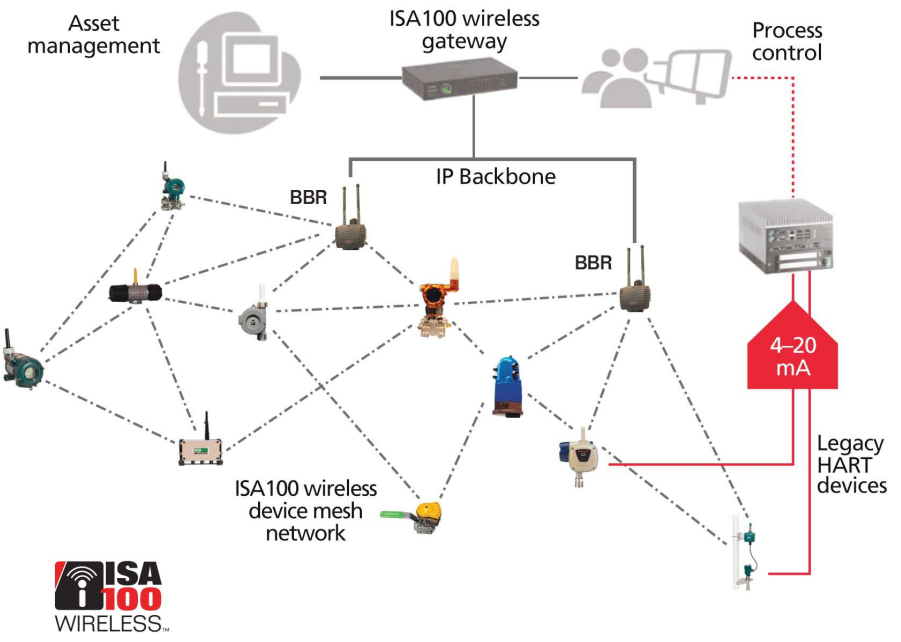


Figure 3. Flexibility of ISA100 Wireless network

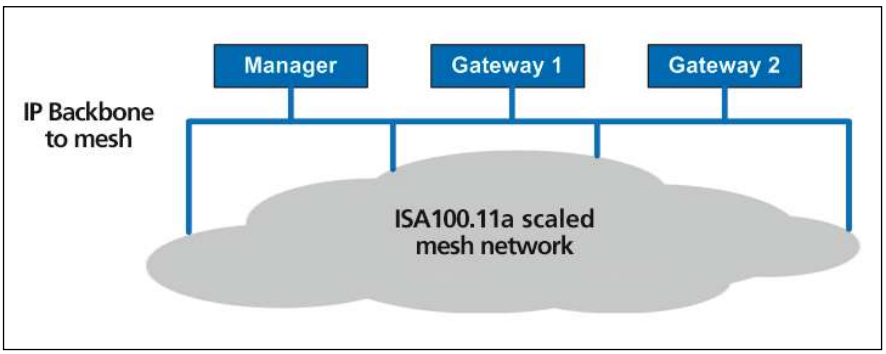


Figure 4. ISA100 structured network design: One network scaled through an IP backbone

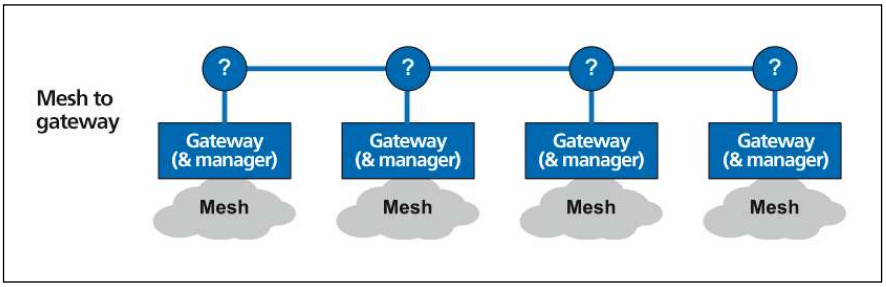


Figure 5. ISA100 ad hoc network design: Multiple small networks scaled by duplication

generation of network managers and automation engineers understand IPv6, and so will future generations.

Figure 3 illustrates the flexibility of the ISA100 Wireless universal network in an example mesh network comprised of devices from multiple manufacturers. Some manufacturers use the native ISA100 Wireless protocol, while others use ISA100 objects with different protocols at the same time. The two legacy HART devices, wired to a controller, are simultaneously reporting HART digital data to the ISA100 Wireless network via ISA100 WirelessHART adapters.

The Internet of Things approach in ISA100 Wireless affects network architecture and the user experience in actual practice.

Figure 4 shows a structured network design that leverages IP. The ISA100 Wireless network is shown as a single cloud or mesh with multiple IP backbone connections. In this configuration, multiple gateways are shown as Internet appliances without a radio. ISA100 Wireless messages flow over the IP network. Networks of this kind are generally rolled out using a structured methodology, similar to the way a Wi-Fi or cellular network is typically deployed.

Figure 5 shows an ad hoc network design. A series of small, typically single-purpose, networks are installed over time. Each network is attached to a device called a gateway. The diagram shows that each gateway also hosts a management function. The connections on the other side of the gateway are shown as a series of question marks, suggesting that the high-side interface is not specified by the standard and could be just about anything. ISA100 Wireless is designed to scale through IP. In practice, small systems or evaluation systems will tend to use ad hoc methodologies as illustrated in figure 5. However, as systems scale, most users will prefer to leverage the IP backbone as illustrated in figure 4. Ad hoc systems tend to be rolled out organically, one project at a time, and can result in unmanageable chaos at scale. For example, unplanned battery failure and communication path instability are typical symptoms of an ad hoc methodology that is inappropriately scaled.

Ready for prime time

Industrial wireless instrumentation is now at a stage where major users are rolling out ISA100 Wireless programs. The "tipping point" for industrial wireless occurs when major users switch from an ad hoc approach to wireless campaigns. In a bottom-up ad hoc approach, small wireless systems are installed one by one to address specific problems, relying on the passion of proactive early adopters. In a top-down campaign, solutions are rolled out to support enterprise strategy. A good process allows for both approaches, with successful ad hoc systems being a laboratory for future wireless campaigns.

Industrial wireless instrumentation is widely considered suitable for monitoring, control, and alarms, including safety alarms. Systems may be deployed using an ad hoc methodology to get started, but users are better served when systems are scaled using a more structured methodology that leverages the Internet of Things. User acceptance of and demand for IP-based scaled mission-critical wireless technology is driving adoption of ISA100 Wireless. ISA100 Wireless is poised to support a fast growth curve with its open design, scalable system and product architectures, and vendor-neutral ecosystem.

For the full white paper, visit www.isa100wci.org/en-US/Learning-Center/White-Papers/ISA100-Wireless-Applications,-Technology,-and-Syst. ■

ABOUT THE AUTHOR

Jay Werb (jayw@isa100wci.org) is the technical director of the ISA100 Wireless Compliance Institute (WCI), where he manages the organization's compliance and other technical programs. He is also the editor and author of the data link layer (mesh) section of the ISA100.11a standard. Werb has more than 30 years of experience in the computer field, with the past 20 years focused on wireless. He has been the technical founder of multiple technology companies and holds more than a dozen patents. In addition to his work with WCI, Werb is a consultant with AIW LLC, where he assists end users with strategic adoption of industrial wireless instrumentation. He has a B.S. in biology and a master's degree in management, both from the Massachusetts Institute of Technology.

View the online version at www.isa.org/intech/20141206.

Protecting your operational integrity

By Clemens Blum



ABOUT THE AUTHOR

Clemens Blum has been Schneider Electric's executive vice president, industry business, since 2010 and is responsible for the integration of Invensys, acquired by Schneider Electric in January 2014. At Schneider Electric, Clemens has served as general manager of the Berger Lahr Group, general manager of the SE Motion Group, and vice president, European division.

New powerful and capable industrial control systems and software solutions have created more opportunities for manufacturers to pursue and achieve greater levels of efficiency, performance, and profitability. Businesses now have more data to measure and analyze, as well as more opportunities to use that data to drive efficiency. This greater interconnectivity between systems and software has also enabled producers to be more agile, particularly in reacting to changing business variables and process conditions.

But these new offerings and capabilities have also created new business vulnerabilities. As manufacturers apply technologies, they must ensure they are not jeopardizing the operational integrity of the plant. Operational integrity is simply the unhindered ability of the system and plant to remain sound and to continue production. In other words, operational integrity means safely and securely mitigating and eliminating threats to business continuity, while meeting or exceeding production targets.

Producers are correctly looking at the promise new technologies bring, namely the ability to use real-time information to better understand their resources, improve how they control costs and business variables, and increase their profitability. The need for real-time operational data to achieve this "promise" has propagated the use of commercial off-the-shelf information technology solutions in industrial environments and shifted the industry toward "connected" network solutions. Now with the Internet of Things, big data, and other emerging trends, connectivity has reached a new level of focus in the discussion, as well as in investments. Because almost everything can be connected to anything from anywhere at any time—at a low cost—new opportunities for improving business processes and performance seem unlimited. For example, at its Rabigh, Saudi Arabia, refinery complex, Rabigh Refining & Petrochemical Company implemented a plant information management system, fully and tightly integrated with its control, SAP, and other production and corporate business applications, to optimize output, improve quality, and increase overall business performance. The solution covers the entire refinery and petrochemical complex comprising 23 plants.

But regardless of what that new technology and better connectivity promise for improving business performance, eliminating and responding to potential risks to operational integrity must continue to be the number one priority. Control systems, especially in the continuous process industries, are critical, not just for driving efficiency and ensuring there is no loss of production, but also for ensuring the safety of the company's assets, people, and environment. Off-the-shelf solutions and higher, more frequent interconnectivity have increasingly exposed industrial control systems to malware and security threats that traditionally target commercial systems. For example, since the Shamoon attack, the preferred target for cybercriminals seems to be the energy sector, where incidents have increased 52 percent since 2012.

Therefore, when deciding when and how to implement or upgrade an industrial control system, the focus cannot be entirely on how newer technology helps achieve production goals. Companies must investigate and understand what and how many layers of protection wrap the system. Those safeguards will enable everyone in a plant to fulfill their roles more effectively. People on the process-connected side of the system will be better able to do their jobs, while those in the control-room side will be able to concentrate on operation performance, without worrying about risks to the integrity of the system.

Secure operating platforms must be delivered with policy, procedure, and layered technical controls to create a "defense-in-depth" model. To handle the new challenges of an even more interconnected world, cybersecurity must be considered first, not as an afterthought. Although the promise of connectivity is great, so is the threat. This is the reason plant and operations managers, especially in critical industries, must ask how any new or upgraded system is protected and what impact the upgrades might have on the security of their operation before they even look at how the solution will increase and improve efficiency.

As industry end users explore and take advantage of new interconnected technology, they can no longer just talk about security. While many threats to plant assets, people, and the environment are external and beyond the plant's control (such as natural disasters), many threats are within the plant's control. Taking active, specific measures to manage technology upgrades and ensure control systems are cybersecure is the first step to protecting the operational integrity of the business. ■

South American steel manufacturer replaces legacy energy management system

By Dave Hellyer



Usiminas is the largest supplier of steel products to South America. With 13 industrial plants strategically located in the states of São Paulo, Minas Gerais, Rio Grande do Sul, Espírito Santo, and Pernambuco, Usiminas produces heavy plate, galvanized steel, electro-galvanized steel, cold-rolled strips, and more for the automotive, civil construction, mechanical industry, road and agriculture machinery, furniture, container, tube, oil and gas, and home appliance markets across Brazil and Latin America.

The Usiminas plant located in Ipatinga, Minas Gerais, Brazil, had been managing an energy system application developed with an older human-machine interface software that was no longer supported by the developer. The energy system collects real-time data from all steel-mill, plate-mill, coking, lamination, continuous casting, and other process units, using advanced optimization algorithms created by Usiminas for performance and demand prediction requirements.

Challenge: Replace legacy system

As part of the energy system, the water monitoring application has a general overview screen and manages 25 water systems in a hot standby configuration. The upgraded system had to support hot standby redundancy for nonstop operations without missing data or experiencing downtime caused by a server failure.

Meeting the challenge required replacing the 100,000 tag legacy system, modernizing the user interface and system architecture, adding more operator stations on the network, and adding a hot standby server to help maximize system availability and uptime. To design, build, and implement its new system, Usiminas collaborated with Tatsoft, which provided consulting, project management, application development, and testing and installation services. Usiminas provided the advanced optimization prediction algorithms as well as committed resources to help implement the system.

One of the most demanding aspects of this project was communicating with serial and Ethernet devices from several different manufacturers simultaneously. To reduce the need for upgrading the controllers and devices on the system, Usiminas leveraged a third-party serial-to-Ethernet interface so industrial devices were directly accessible from the network. Legacy devices can be monitored and controlled from any network location or the Internet. Different configurations and features are available for specific applications, such as protocol conversion, real COM drivers, and TCP operation modes. OPC servers were used for several other devices. Usiminas also had to interface with Oracle and Microsoft SQL databases where historical alarms and trend and process data are stored and managed using the connectivity tools within the new system.

The old system required 100,000 tags. It has been replaced with new software that requires less than half that number of tags, due to the built-in properties and application objects. The new system now includes 35 nodes distributed across the entire plant. Operators can monitor the unit operations with remote clients, so they are not tied to a single desktop.

To help ensure the correct operators are logged in during a shift, the software has an automatic logout feature. After eight hours, the software logs out the operator and gives access only to the operator registered in the current shift. Another feature is a startup delay, which prevents the system from generating erroneous alarm conditions as it starts up.

Native features in the new software notify the operator if an active alarm requiring action has not been resolved after a specific amount of time—helping avoid human failure. The system controls external components, including third-party Windows services, and other external processes for greater stability when the primary station unexpectedly shuts down.

The displays show active screen images of pumps, motors, and valves, and the operator can add comments about operations, which other operators can read to know what is happening or for specific process details. A summary of actions that includes those comments is kept.

The security system can manage shifts of operators. The currently logged-in operator can execute only the specific actions for which he or she has clearance. In these cases, the operator can only log into the system if he or she is on the correct shift. There are three shifts configured, with a specific group of operators in each. The system executes an "auto-logoff" at the end of the shift to be ready for the next operator on the next shift.

One final major benefit for Usiminas is that the new software fully leverages the resources available from the computer, including memory, multicore CPUs, and advanced graphics cards. To take full advantage of those resources, the computer used as the server for this system includes Windows 64-bit OS and 10 GB RAM.

Increased reliability, efficiency

Guilherme Publio Teixeira, manager of the energy and utilities systems support group at the Usiminas plant, said, "… software, used in the migration of the integrated recirculation monitoring system at the Ipatinga facility proved to be stable and reliable, increasing the process efficiency and cost optimization." ■

ABOUT THE AUTHOR

Dave Hellyer (david.hellyer@tatsoft.com) is vice president of sales and marketing at Tatsoft, a member of the Control System Integrators Association.

In memoriam

Ed Sullivan



ISA instructor Ed Sullivan passed away on 11 October 2014. Sullivan was a veteran instrumentation professional who began his career in the U.S. Air Force. He was an instrumentation technician, a design engineer in both the office and the field, and an instructor in the classroom and plant. He worked in education, agriculture, aviation (licensed commercial pilot), chemical, food, medical, pharmaceutical, and textiles for organizations including Allied Chemical Corporation, Toledo Scales Distributors, Lockwood Greene Engineers, and the U.S. government. He had an extensive background in instrumentation and electrical design, application, documentation, and field startups. Sullivan was a member of the National Occupational Competency Testing Institute and the author of numerous calibration documentation procedures. He also developed calibration data formats and established preventative maintenance programs.

Sullivan started teaching for ISA in 1999 with the objective "to share my knowledge and experience with those who are to follow me." He will be sorely missed by members of his ISA family, but remembered as a valuable colleague and individual to have known and worked with. He contributed much of his knowledge and expertise to the success of ISA courses. He challenged his students to think beyond what was on paper or right in front of them.

Keith Otto



Former chair of the *InTech* editorial advisory board, Keith David Otto, P.E., passed away on 14 October 2014. After serving in the U.S. Navy as a gunfire control technician and earning a B.S. in electrical engineering, Otto worked on the Thor and Titan missile/rocket projects early in his career at AC Spark Plug. His career also included positions at American Can Co., Fischer & Porter Co., and Marathon Engineering, where he also served a 13-year term as president. Otto went on to work for James River Corporation as director of manufacturing technology and later cofounded Neenah Engineering Inc., where he worked until May 2014.

He was an active member of several technical societies, including ISA, TAPPI, IEEE, IAAI, and IAEI. Otto is the namesake of the Keith Otto Award, which is presented each year to the author of the best article in *InTech*. Otto was a lifelong member of ISA who served on the editorial advisory board. Colleague Walt Boyes remarked, "Keith was always a voice of professionalism and moderation on the publications board and on the *InTech* editorial advisory board. He was one of those fantastically competent, quiet engineers who was much more interested in getting things done than in self-aggrandizement." He will be truly missed.

Jerry Voss



ISA67 member and ISA instructor Jerry Voss passed away on 1 November 2014. Voss was a highly qualified instrumentation and control systems specialist with many years of experience in U.S. naval and commercial nuclear reactor facilities. He specialized in project management and programmatic/design support in the areas of software quality assurance, digital process improvement, set point analysis, drift analysis, and licensing of instrumentation.

Voss was a longtime POWID Symposium supporter, presenter, and session chair. He was also a valued ISA course instructor, course developer, and standards leader. Colleague Ron Jarrett wrote that "we have lost a great leader and more importantly a dear friend." His family explained that he really much preferred to be working toward a goal, getting things done, and helping someone than having all eyes on him. He was passionate about getting things right. Voss cared deeply about his colleagues, friends, and family and was willing to go out of his way to help out, share a laugh, or just listen to someone's concerns. In his professional life, he strived to give of his best to clients and colleagues alike.

2015 ISA division symposia

2015 Food and Pharmaceutical Industries Division (FPID)
9–12 February 2015
Siemens Industry, Inc.
Philadelphia, Pa.

60th ISA Analysis Division Symposium (AD)
26–30 April 2015
Galveston Convention Center
Galveston, Texas

15th ISA LDAR-Fugitive Emissions Symposium
19–21 May 2015
Astor Crowne Plaza
New Orleans, La.

61st International Instrumentation Symposium (IIS)
Co-sponsored by MFPT
11–15 May 2015
Westin Hotel
Huntsville, Ala.

58th Power Industry Division Symposium (POWID)
7–11 June 2015
Kansas City Marriott
Kansas City, Mo.

2015 Water/Wastewater and Automatic Controls (WWAC)
4–6 August 2015
Wyndham Lake Buena Vista Resort
Orlando, Fla.

2015 Process Control and Safety Symposia
9–12 November 2015 (tentative)
Norris Convention Center
Houston, Texas

ISA Certified Automation Professional (CAP) program

CAP question

Consider the following automation system data:

- Preventive maintenance for 1 hour every month
- Quarterly preventive maintenance for 2 hours each quarter
- One failure that results in 6 hours of downtime
- One failure that results in 4 hours of downtime

What is the uptime for this automation system if it runs 24 hours a day, 365 days a year?

A. 99.66%
B. 99.77%
C. 99.86%
D. 99.89%

CAP answer

The most important measure for production equipment support is operational availability, or "uptime."

Automation equipment that operates for 365 days x 24 hours per day = 8,760 total possible "up" hours. This equipment gets preventive maintenance for 1 hour every month (12 hours per year), plus additional quarterly preventive maintenance of another 2 hours each quarter (8 more hours per year).

There was one failure that resulted in 6 hours of downtime and a second failure that resulted in 4 hours of downtime. Thus, total downtime for all maintenance was 12 + 8 + 6 + 4 = 30.

The correct answer is *A*, "99.66%."

$$\textit{Operational Availability (Ao)} = \frac{\textit{Uptime (Operational Time)}}{\textit{Total Time}} = \frac{8760 - 30}{8760} = 99.66\%$$

Reference: Trevathan, Vernon L., *A Guide to the Automation Body of Knowledge, Second Edition,* ISA, 2006.

Certified Automation Professionals (CAPs) are responsible for the direction, design, and deployment of systems and equipment for manufacturing and control systems.

ISA Certified Control Systems Technician (CCST) program

CCST question

Which of the following is a valid form of the PID algorithm?

A. Proportional and derivative modes on error, integral on measurement
B. Proportional and integral modes on measurement, derivative on error
C. Proportional, integral, and derivative modes on error
D. Proportional, integral, and derivative modes on measurement

CCST answer

The correct answer is *C*, "Proportional, integral, and derivative modes on error."

The standard PID algorithm is configured so that each of the three terms operates on the error (SP – PV). Proportional and derivative actions can also be configured to operate on PV (measurement), but not integral. Integral requires a reference against which to integrate (i.e., SP as a reference, and error as the quantity over which to integrate).

Answers A, B, and D all include "integral on measurement," and therefore are not valid forms of the PID algorithm.

Reference: Goettsche, L. D. (Editor), *Maintenance of Instruments and Systems, Second Edition,* ISA, 2005.

Certified Control System Technicians (CCSTs) calibrate, document, troubleshoot, and repair/replace instrumentation for systems that measure and control level, temperature, pressure, flow, and other process variables.

Distillation column loop tuning

Trust the operator, but verify the basics

By Harley Jeffery

Controlling distillation columns is a tough assignment due to the interacting nature of the process and the upstream/downstream effects of loop tuning. In this example, the column bottoms level control had stability problems (cycling), causing manual operation and the operators' constant attention.

The bottoms level needed to be controlled to a set point that minimized the time the liquid sat in the receiver for quality purposes. However, it had to maintain a fairly tight range to prevent flooding the bottom tray on the range upper end and to maintain sufficient pump suction head for the bottoms pump out for the lower constraint.

Previous loop tuning had resulted in excessive movements to the bottoms flow that adversely loaded the downstream column. Thus, the operators ran the level-to-flow cascade control loops in manual and constantly manipulated the outflow to maintain the bottoms level within the desired range. Of course, as other matters required operator attention, the bottoms level would drift. When the operator intervened for the required correction, the resultant upset of the columns affected production rate and quality.

Discussions with operating personnel revealed that they typically let the level float within a "comfort" range, because the process would fluctuate but not to the point of concern. If it looked like an up or down trend was developing, then a small bump was made to the outlet flow, and they waited for correction. If the deviation from set point became greater, then larger bumps to the outflow were performed, and they again waited for correction. When asked to place the level/flow in cascade operating mode, the level loop would typically overcorrect and begin to cycle. The outlet flow loop was considered to be working well even though it was manipulated in manual.

The operators had learned from experience what appeared to be a nonlinear gain control strategy. Trusting their judgment, we then proceeded to verify the basics before implementing the strategy.

EMERSON Process Management — CONTROL SOUTHERN

Loop Inspection Data

Customer:				
Plant/Mill:				
Area:	Distillation			
Loop:	LIC-XXX		Field Engineer:	HMJ
Description:	Bottoms Level Control		Date:	25-Apr-13
Controller				
Manufacturer:	Emerson DeltaV	Proportional:	0.400	Gain
Algorithm:	Std PID	Integral:	160.00	Sec/repeat
Scan Rate:	2 sec	Derivative:	0.00	Sec
Engr. Units:	0-108 inches	PV Filter:	5.000	Sec
Notes: Mode= Auto/IMAN; ARW Lo=0% and ARW Hi=100%/OUT_SCALE=0-4500 lb/hr				
ISEL for LI-32002 and LI-32004 - 5 sec execution				
Measurement				
Manufacturer:	Magnetrol	Model:	705-520A-110	Digital bus
Meas. Principle:	Radar	Range/Calibratn:	0-108	in H2O
Protocol (Smart):	Foundation fieldbus	Output:	0-108	inches/Ff
I/O Location:		Dampening:	--	Sec/Min
Installation Notes:				
Final Control Element				
Cascade Master to FIC-XXXXX: Feed Flow to Downstream Column				
Manufacturer:		Model:		
Serial Number:		Style:		
Travel:		Shutoff Class:		
Size:		Characteristic:		
Actuator				
Manufacturer:		Model:		
Serial Number:		Style:		
Air Pressure:		Bench Set:		
Positioner				
Manufacturer:		Model:		
Serial Number:		Input Signal:		
I/P				
Manufacturer:		Model:		
Serial Number:		Input Signal:		
I/O Location:				
Installation Notes:				

EMERSON Process Management — CONTROL SOUTHERN

Loop Inspection Data

Customer:				
Plant/Mill:				
Area:	Distillation			
Loop:	FIC-YYY		Field Engineer:	HMJ
Description:	Feed to Downstream Column Flow Con		Date:	25-Apr-13
Controller				
Manufacturer:	Emerson DeltaV	Proportional:	0.150	Gain
Algorithm:	Std PID	Integral:	25.00	Sec/repeat
Scan Rate:	1 sec	Derivative:	15.00	Sec
Engr. Units:	0-4500 lb/hr	PV Filter:	3.000	Sec
Notes: Mode= AUTO				
Measurement				
Manufacturer:	Micro Motion	Model:	2700C	Analog
Meas. Principle:	Coriolis	Range/Calibratn:	0-4500	lb/hr
Protocol (Smart):	Foundation fieldbus	Output:	0-4500	lb/hr - Ff
I/O Location:		Dampening:	--	Sec/Min
Installation Notes:				
Final Control Element				
Manufacturer:	Fisher	Model:	EZ	
Serial Number:	16668824	Style:	Globe	
Travel:	3/4 in	Shutoff Class:	IV	
Size:	1 inch	Characteristic:	Equal Percentage (microform)	
Actuator				
Manufacturer:	Fisher	Model:	667	
Serial Number:	16668824	Style:	Spring & Diaphragm	
Air Pressure:		Bench Set:	14-30 psi	
Positioner				
Manufacturer:	Fisher	Model:	DVC5010F	
Serial Number:	16668824	Input Signal:	0-100% - Foundation fieldbus	
I/P				
Manufacturer:		Model:		
Serial Number:		Input Signal:		
I/O Location:				
Installation Notes:				

Figure 1. Loop inspection data forms

Basics – Loop tuning and performance

Plant walk through

The first step for loop performance benchmarking is walking through the process to investigate the type and installation of the process measurement and final control elements. Then, returning to the control room, we recorded the distributed control system (DCS) control strategy and tuning. The loop inspection data forms were used to document the as-found control (figure 1). The measurements were selected with the latest technologies and appeared to be installed with good practices. The control valves were a high-performance design and offered no constraints to push loop response if needed. The as-found tuning parameters were questionable and therefore confirmed our process of "verifying" base-level loop performance.

The DCS configuration application was traced to verify a typical level-to-flow cascade strategy per figure 2. The DCS also has several built-in cascade application enhancements that we activated.

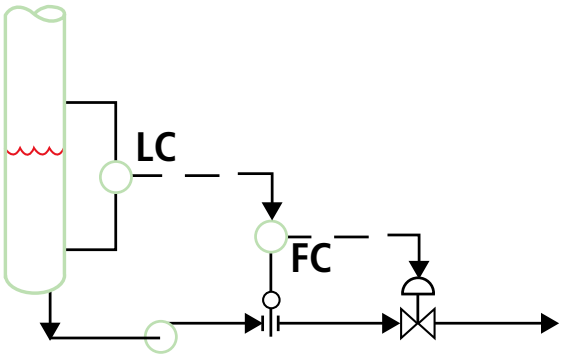


Figure 2. Bottoms level-to-flow cascade strategy

Designing and testing

Verifying performance begins with designing the loop tests, which include gathering time series data of the process measurements to analyze the overall process trends, statistics for variability, and response to upsets. Then we perform individual loop bump tests to gain process dynamic data and valve performance. We elected to use the EnTech Toolkit to collect, analyze, and help tune the control loops. However, this plant uses FOUNDATION Fieldbus, which communicates digitally to the DCS. The toolkit requires a voltage signal for the process inputs. A recently added feature of the DCS uses characterizable I/O modules, so they were employed to read the process variables over the DCS communications structure and reproduce the values on analog output (AO) modules (figure 3). The toolkit was then connected to these AOs and successfully able to collect pertinent process data from the DCS.



Figure 3. Collecting analog data from FOUNDATION fieldbus devices

With the toolkit gathering process data, we could collect time series short-term data and overnight runs for longer-term data and saw the disturbance to the level and various operator responses. The loop bump testing confirmed that the slave flow loop was capable of aggressive tuning with good response to 0.5 percent bumps (figure 4).

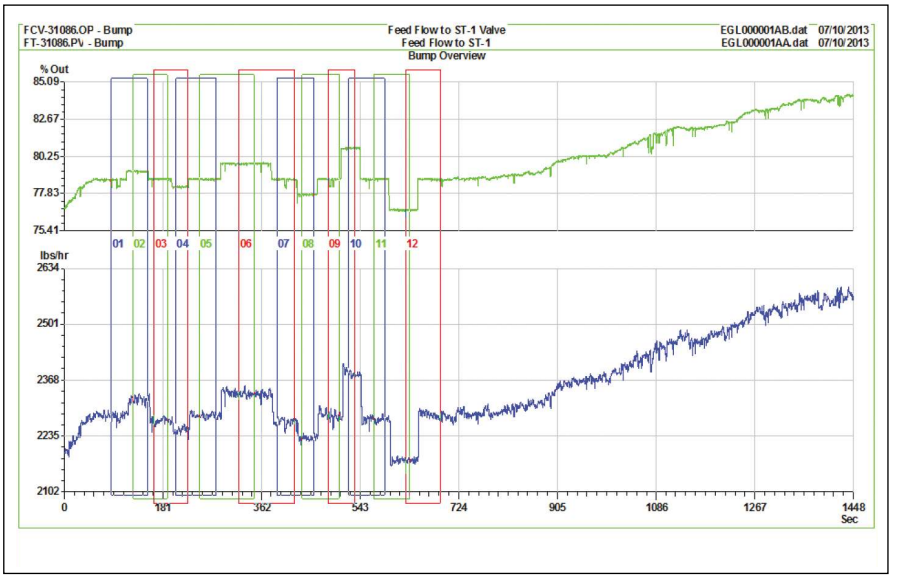


Figure 4. Flow loop bump test

Loop tuning and verifying performance

The flow loop was retuned with the new parameters to minimize delay in response to the master level control. With the flow loop in cascade mode, the level output bumps verified an integrating process with a long dead time. The level controller was retuned with increased gain and very slow reset. With the revised tuning, the level

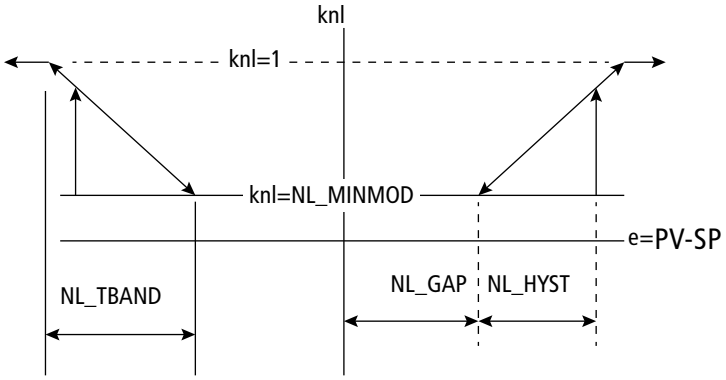


Figure 5. Nonlinear gain algorithm (NLG)

controller was then put in automatic with the flow loop in cascade. We monitored the performance. The results showed cycling was minimized, and the set point was maintained within the desired limits.

However, the standard proportional, integral, derivative (PID) level-to-flow cascade manipulated the flow to maintain the set point with the same response for small deviations, as well as larger ones. This degree of flow change to the downstream movement caused disturbances to the highly interactive distillation process.



Advanced control (trusting the operator)

The question then became how to maintain an acceptable bottoms level while minimizing downstream disturbances caused by outlet flow changes. Returning to the operator's method—manually making small changes to the outlet flow to maintain the level within a "comfort" band around the set point and only making larger flow changes if the level was approaching a constraint—had the benefit of minimizing flow disturbances to the downstream column.

The operator method looked like a good example of the nonlinear gain control algorithm (figure 5). This technique uses a "gap" around the set point where small gain is in effect. This is equivalent to the operator not changing the outlet flow if the level is within his or her comfort zone. However, as the PV-SP error increases, the gain is increased to make the appropriate correction; again, as the operator sees that the level is, in fact, headed to a constraint, then additional outlet flow is used to correct the level.

The DCS has a built-in NLG function that can be "enabled" on the PID algorithm:

- *NL_MINMOD* is the gain applied when the absolute value of the error is less than NL_GAP. To get deadband behavior, set NL_MINMOD to 0.
- *NL_GAP* is the control gap. When the absolute value of the error is less than NL_GAP, KNL = NL_MINMOD.
- *NL_TBAND* is the transition band over which KNL is linearly adjusted as a function of error.
- *NL_HYST* is a hysteresis value. Until the absolute value of the error exceeds NL_GAP + NL_HYST, KNL = NL_MINMOD. Once the absolute value of the error has exceeded NL_GAP + NL_HYST, the absolute value of error must return to a value less than NL_GAP before KNL returns to a value of NL_MINMOD. If NL_GAP is 0, then the value of NL_HYST has no meaning (effectively assumed to be 0).

We then enabled this feature and collected additional data/observations in order to tune the gap, transition band, min mod (gain), and hysteresis. In general, we tried to mimic the operator's method and add a smoother implementation of the flow set point changes.

Results

Placing the bottoms level "master" controller to automatic and the "slave" outlet flow controller to cascade with the nonlinear gain enabled and tuned, the level was maintained with acceptable limits around set point while minimizing the flow to the downstream distillation process. Recent follow-up with the plant site indicated that the loops are still in automatic, allowing operators to spend time in more productive efforts. ■

ABOUT THE AUTHOR

Harley Jeffery (harley.jeffery@controlsouthern.com) has worked in the field of industrial process control for more than 40 years, starting with Fisher Controls as a valve application engineer. He has been a control systems engineer and engineering manager with Control Southern for the past 34 years. His experience includes design, implementation, testing, startup, and ongoing improvement of DCSs. Jeffery received a BSIE from Louisiana State University.

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Who owns your career?

By Dean Ford, CAP

Because previous columns have focused on the intricacies of developing the workforce of the future, I want to take a slightly different twist on this important topic. From primary education through graduate learning, workforce development is vital to our growth as a profession. ISA has formed a partnership with FIRST (For Inspiration and Recognition of Science and Technology), and the Automation Federation has developed the Automation Competency Model and is working with various learning institutions to develop automation-specific curriculum. With all of this work going on, what are we, the professionals who currently make up the automation workforce, doing to enhance our careers and support our own profession?

What is your career path?

Over the years, I have seen that an overwhelming majority of automation professionals do not have a solid, actionable career plan. I cannot explain why this is the case, but it seems to be the reality. How many of us have determined what we want from our careers? What do we want to be in five, 10, 15 years? The norm is the passive path, where we put our careers in the hands of others. As we gain more seniority in a company, often we are expected to take on more responsibility or we are placed in a position where our expertise and passion are misplaced. For instance, a great senior technical person may be placed into a project management role in which he or she will likely struggle to be successful and will feel out of place.

Automation professionals can find themselves in a variety of roles, such as technicians, engineers, managers, sales, service, field techs, commissioning specialists, technical gurus, and process experts. We can work for end users, systems integrators, vendors, engineering firms, ourselves, and many other employers. The possibilities are endless. With all these options, how do you differentiate yourself from others? How do you demonstrate your mastery of automation?

You have to be an expert at something

To succeed, you have to be an expert at something and be able to demonstrate that expertise to others. It is difficult to be an expert in the total field of automation, because the body of knowledge is so expansive. One of the best ways to demonstrate your mastery of the profession is that of certification. ISA has set the standard for certification programs in the automation profession. The Certified Control Systems Technician (CCST) and Certified Automation Professional (CAP) are designed specifically for this. CAP is a globally recognized certification demonstrating that the holder has a working knowledge of the automation profession and can be trusted to perform the duties required in automation.

Another avenue for U.S.-based automation professionals is the Control Systems Engineer (CSE) Professional Engineer's license. Licensure, although not officially globally recognized, is still a strong differentiator for automation professionals. In addition, the CSE allows engineers to seal designs in the states they are licensed. The CCST and CAP are unique in that they are globally recognized. The CSE is only a state-by-state recognition in the U.S. if you want to use it in its official capacity.

To succeed, you have to be an expert at something and be able to demonstrate that expertise to others.

None of these programs should be taken lightly. They come with obligations to perform your job ethically and professionally. The exams are tough and rightly so. To obtain the certification, it takes preparation and a lifelong commitment from you. There are ongoing requirements to keep your certification active and relevant. Once you obtain the certification, you too are held to the higher standard that the certification demands.

These certifications are administered by third parties and designed with very strict standards. They can be used by you to demonstrate to your employer and clients that you are an expert in your profession. It is important to note that by obtaining these certifications, you are committing yourself to upholding the high standards of the profession and to performing your job accordingly. Your peers, employers, and clients know they can trust you to do the right thing.

Take charge!

You own your career. Do not put it in the hands of your employer or manager. Get involved in your profession. Differentiate yourself from others by obtaining recognition through certification, licensure, or both. When you take active ownership of your career, you will see a marked improvement in your job satisfaction. Do your research and make sure the path you choose is the one you want. Be prepared to make the necessary changes to reach your goals.

I look forward to welcoming you to the group of certified professionals who have taken charge of their careers. ■

ABOUT THE AUTHOR

Dean Ford, CAP, (Dean.ford@we-inc.com) serves the professionals of Westin Engineering, Inc., as VP North American Engineering. In this role, he keeps our electrical, water and wastewater infrastructure safe, secure, and reliable—protecting both the public and the environment.

Reference: Trevathan, Vernon L., *A Guide to the Automation Body of Knowledge, Second Edition,* ISA, 2006.

Alarm management update

ANSI/ISA-18.2-2009, *Management of Alarm Systems for the Process Industries,* has found wide use across the process industry sectors. The standard addresses the development, design, installation, and management of alarm systems in the process industries. Alarm system management includes multiple work processes throughout the alarm system life cycle. The standard defines the terminology and models to develop an alarm system and the work processes recommended to effectively maintain the alarm system throughout the life cycle.

The IEC version of the standard, IEC 62682, has been completed and will be published by the end of 2014. ISA18 co-chairs Donald Dunn and Nick Sands led the IEC work as convener and secretary/editor, respectively. Dunn is the director of engineering in the Amarillo Division for Phillips 66. Sands is the global alarm management leader for DuPont and a Manufacturing Technology Fellow for DuPont Protection Technologies.

InTech caught up with the two at the ISA Leaders Meeting in Kansas City, Mo., in early November for a brief interview following a one-day meeting of ISA18.

Are there any significant changes in the forthcoming IEC version from the original ISA standard?

Dunn/Sands: IEC 62682 eliminates country-centric criteria that were present in the original standard. In addition, there was a shift of the normative criteria ("should" to "shall") within IEC 62682. The IEC work utilized one editor (Sands) of the document in lieu of the clause editors used in writing the original ISA-18.2, promoting a more cohesive document with consistent wording and terminology throughout.

What are the key goals for ISA18 in 2015?

Dunn/Sands: The bulk of the subject-matter experts on alarm management participate on ISA18. The starting point in the IEC work, of course, was ANSI/ISA-18.2-2009. As co-chairs of ISA18, we elected to edit the 2009 standard by incorporating the changes implemented during the IEC work and then submitting the document to the ISA18 committee for comments. Thus, ISA18 is now in a very good position to complete a revision of ISA-18.2 in the coming year. Other goals for ISA18 in 2015 include publishing additional ISA technical reports (TR) to provide support in using the standard, covering *Identification and Rationalization* (TR2) and *Basic Alarm Design* (TR3); and nearing completion of two additional technical reports, *Philosophy* (TR1) and *Alarm Management for Packaged Systems* (TR7). We also expect progress on the revision of ISA-18.1, *Annunciator Sequences and Specifications*.

The standard defines the terminology and models to develop an alarm system and the work processes recommended to effectively maintain the alarm system throughout the life cycle.

The ongoing ISA18 work cited by Sands and Dunn reflects a continuation of the development of a comprehensive series of technical reports to support the standard. Previously published technical reports in the series include:

- ISA-TR18.2.5-2012, *Alarm System Monitoring, Assessment, and Auditing,* provides guidance and information supplementing the standard on the use of alarm system analysis for both ongoing monitoring and periodic performance assessment. Monitoring, assessment, and audit are essential to achieving and maintaining the performance objectives of the alarm system. These activities can identify improvement opportunities in the other life-cycle stages, such as philosophy, rationalization, detailed design, implementation, operation, maintenance, and management of change.
- ISA-TR18.2.4-2012, *Enhanced and Advanced Alarm Methods,* helps users evaluate when to use enhanced and advanced alarming methods, what benefits they can achieve, and what challenges and costs to expect. Per ANSI/ISA-18.2-2009, enhanced and advanced alarm methods typically go beyond the basic methods and techniques that are usually, or at least initially, applied. Although significant improvement in alarm system function and performance can usually be made by following the basic alarming methods and principles, in some cases they may not be sufficient to achieve the goals for performance and operator guidance stated in the alarm philosophy.
- ISA-TR18.2.6-2012, *Alarm Systems for Batch and Discrete Processes,* covers the application of alarm management principles in the standard to batch and discrete processes. The general principles and techniques described are intended for use in the life-cycle management of an alarm system based on programmable electronic controller and computer-based human-machine interface technology. Use of the technical report should consider batch and discrete process alarms from all systems presented to the operator, which may include basic process control systems, annunciator panels, safety instrumented systems, fire and gas systems, and emergency response systems. Following the recommended guidance will help to identify and address alarm specification, design, implementation, and management opportunities that are important to batch and discrete processes. It can also help minimize the generation of nuisance alarms that could complicate and frustrate an operator's awareness, understanding, and response to abnormal situations.

For information about viewing or obtaining the ANSI/ISA-18.2-2009 standard and the supporting technical reports, visit www.isa.org/standards. For information about ISA18, contact Charley Robinson, crobinson@isa.org. ■

Focus on loop tuning

Control performance analytics



The Control Performance Analytics service for the process industry includes the Simatic PCS 7 process control system, which captures process and status data from control loops. The service is suitable for the chemical and pharmaceutical, food production, oil and gas, and glass and solar industries. Like the "plug-and-play" principle, the service can be quickly installed during ongoing operation with a relatively small financial investment. A data collector, such as a nanobox, is installed on site. This device captures process data from control loops (e.g., process tags and alarms) previously defined by the user. It then transmits the data to the Siemens operation center. The experts there analyze the data and make their evaluations available to the customer as reports via a web portal.

Siemens, www.siemens.com

Thermal control solutions

The customizable, application-specific thermal control system has an EZ-ZONE RM temperature controller. The controller is flexible and scalable; users can mix inputs and outputs to configure control loops, limit loops, and monitoring points. The solution improves system reliability, eliminates compatibility issues, and reduces wiring time and complexity. It also has full EtherCAT communication protocols via the Semiconductor Temperature Controller Profile ETG.5003.2060 specified by the EtherCAT Technology Group. The solution also has power switching capable of 2 amps to 15 amps internally or pilot duty switching of high amperage systems.

A touch screen operator interface terminal allows an operator to monitor or make enhancements to the process, monitor alarms, and communicate various parameters to multiple devices. Users can create each individual zone of control on the screen. With a touch to the screen, operators can change control and alert parameters, view data, and evaluate historical events. The software, EZ Builder 500, allows users to customize the look and feel of the system to match their company's brand.

Watlow, www.watlow.com

Temperature control I/O



The UIS-04PTN PT100 temperature control I/O module may be either snapped onto the back of a UniStream human-machine interface (HMI) panel next to a CPU to create an all-in-one HMI + programmable logic controller (PLC), or installed on a standard DIN rail using a local expansion adapter to avoid wiring to the door. It combines dual CPUs, a variety of HMI touch panels, and local and remote I/O installation to help original equipment manufacturers and system integrators cut costs, save space, and reduce programming time.

Users can remotely access the module through PCs, tablets, and smartphones via VNC; they can also cut system programming time by 50 percent, because the system anticipates the programmer's intentions and allows them to reuse written code.

Unitronics, www.unitronics.com

Temperature controllers



The CN38S series of temperature controllers is easy to program and has an input for thermocouple, RTD, and thermistor; a sensitive touch keypad; and programmable password protection. The sensor keyboard guarantees protection from dust and liquids in every critical environmental situation. The soft-start makes it possible to eliminate thermal shock and mechanical stress during start up. It is suitable for the process manufacturing, chemical, automotive, and water industries.

Omega, www.omega.com

PXU PID controllers

The PXU series of proportional-integral-derivative controllers is designed for machine builders, systems integrators, and commercial equipment manufacturers. The series enables tighter, more reliable control over a wide range of processes, including temperature, flow, and pressure, from a single model.

Capable of being deployed in precision process applications within the food and beverage, plastics, packaging, energy, gas, heat-treating, commercial, and medical industries, the PXU series offers 1/16, 1/8 and 1/4 DIN-size models. The series has a bezel design with large displays for better visibility from long distances, optional RS-485 communications for integration with PCs, PLCs, and HMIs, and 1-inch shorter depth to save panel space. It is programmable via the front panel push buttons or the company's Crimson configuration software, and it has on-demand auto-tuning. Universal inputs support thermocouple, RTD, 0–10 VDC, and 4–20 mA (0–50).

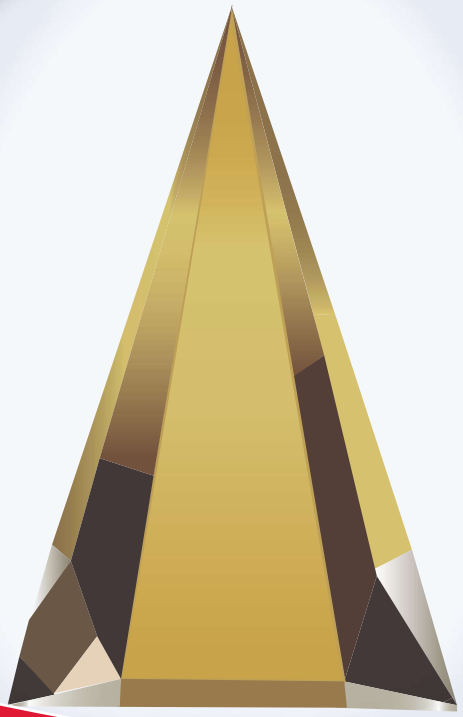
Red Lion, www.redlion.net

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ISA Process Management and Control Division
A. Vimala Juliet, Ph.D.
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Welcome to the next generation of automation media!

By Rick Zabel



ABOUT THE AUTHOR

Rick Zabel (rick@automation.com) is vice president, publisher, and occasional editor of Automation.com, and has been with the company since its inception in 2000. Previously, Zabel worked with Wunderlich-Malec Engineering Inc. of Minnetonka, Minn., where he was the marketing manager for the process control and software integration business groups. Zabel has a B.S. in electrical engineering.

We all live and work in a global, competitive environment. It is an environment that requires companies to look internally and evaluate their strengths and weaknesses. It is an environment that compels companies to look externally for complements to their existing product or service offerings. It is an environment that enables potential business partnerships and possible acquisitions. This environment spawned a strategic publishing partnership more than five years ago between two automation-focused organizations: ISA and Automation.com. The natural progression of that partnership was the recent acquisition of Automation.com by ISA.

The International Society of Automation (ISA) and Automation.com began their partnership in October of 2009. At that time, ISA published the well-established *InTech* magazine, as well as *InTech* online and *InTech* e-newsletters. ISA wanted to continue and expand their electronic content offering to their members and subscribers. To do that effectively and efficiently, the organization needed a partner.

Since the company formed in 2000, Automation.com developed as a proven, profitable, digital-only media company. Automation.com became the leading automation-related online news and resource site and the publisher of *Automation Weekly* and various topical e-newsletters. ISA recognized the strength of Automation.com's e-newsletter offering. Because we operated in the same market, the two companies partnered to offer a comprehensive series of electronic newsletters to their combined automation audiences. In 2009, the distribution list of the newly co-branded newsletters was 90,000. Today, our newsletters are distributed to 140,000 automation professionals worldwide. The partnership benefited both ISA and Automation.com.

Over the past five years, our two companies learned a lot from each other, particularly each other's strengths. ISA, with approximately 30,000 members, excels at generating relevant technical content for automation standards, training, symposia, books, and *InTech* magazine; and ISA has developed a renowned corporate sponsorship program. Automation.com's strength is content delivery and offering exciting, effective advertising opportunities in digital channels.

ISA is a healthy, well-established, individual member-based organization. Over the decades, ISA established and grew a significant investment portfolio. In the past two years, it made significant website and infrastructure investments to enable faster growth. ISA saw the opportunity to take our partnership to a new level by purchasing Automation.com.

ISA leadership began talking internally about the possibility of purchasing Automation.com one year ago. In March of this year, those conversations turned external to include the equity partners of Automation.com. When I was presented with the acquisition possibility, I immediately voiced my support. I have led the Automation.com media business since its inception in 2000. I understand ISA, the automation and publishing markets, and their dynamics. I have seen the positive effects our partnership had on our media products, readers, and advertisers. I immediately recognized the potential collaborative synergies that could occur if we became united. In my opinion, it was the proverbial "no-brainer" progression of our relationship. I think everyone will agree that it makes perfect sense for ISA to own Automation.com.

ISA purposefully created Automation.com as an independent, for-profit company to sustain its entrepreneurial, nimble spirit. It is important to understand that Automation.com will continue to operate as it has for the past fifteen years. ISA and Automation.com will now be able to present a more cohesive publishing strategy to benefit readers, advertisers, ISA's partners, and the automation community as a whole. Our now collective, integrated media offering includes the Automation.com website, *InTech* website, co-branded e-newsletters, *InTech* print magazine, and the new *InTech Plus* mobile app and *PULSE* digital magazine.

The acquisition will allow us to explore other possible publishing activities and new content delivery options. One of ISA's strategic goals is to provide automation professionals with the "coolest delivery" of content. This acquisition will help ISA achieve that goal. Together we have the opportunity to become the official voice of the automation profession.

One thing I have learned in my career is the value of networks and teamwork. A collection of companies or individuals will always be more powerful and more successful than one single company or individual. The combination of ISA and Automation.com will be greater than the sum of the two individual organizations. There are endless possibilities and opportunities yet to be explored. Please send me your feedback. I look forward to being a part of the ISA family and the entire automation community. ■

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